



UNIVERSITY OF ENERGY AND NATURAL RESOURCES, SUNYANI

SCHOOL OF SCIENCES

DEPARTMENT OF INFORMATION TECHNOLOGY AND DECISION SCIENCES

**AI POWERED FOOD RECIPE GENERATOR FROM FOOD IMAGES WITH MOBILE
APPLICATION INTEGRATION**

BY

UEB3222323 - TETTEH MOSES

UEB3207321 - ASIAW JEFF

UEB3205921 - AGYEIBEA SUSANA DANSO

UEB3213421 - OTU DERRICK

UEB3224221 - ADU MENSAH DERRICK

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DECLARATION

Agyeibea Susana Danso, Adu Mensah Derrick, Otu Derrick, Asiaw Jeff, And Tetteh Moses declare that this submission is our effort toward earning a Bsc. in Information Technology at the University of Energy and Natural Resources and that, to the best of our knowledge, it does not contain any material that has already been published by another person or material that has been accepted for the awards of any other qualification of the university, with the exception of where proper acknowledgment has been made.

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Abstract

Alternative methods such as text-based keyword search methods of recipe discovery are not always easy with those users who may not know a dish by name or ingredients. This project tries to help mitigate this limitation, presenting a smart food recipe development tool that will process actual food pictures through an intelligent neural network to generate cooking guides and lists of ingredients. The system involves the multimodal approach where visual features are extracted by Convolutional Neural Network (CNN) of a food image whereas Transformer model is used to create a cohesive text recipe. The system has been designed to use the Recipe1M+ dataset to address major weaknesses such as the semantic gap in all forms of visual and textual data as well as the differences in food presentation, lighting as well as camera angle. The crux of the system is based on a strong architecture comprising of an Image Processing Subsystem and a Recipe Generation Subsystem which run on a user-friendly mobile application, written in Flutter with the backend being a Django framework. The idea of the project is to make recipe discovery more efficient and accessible and provide a service that could be valuable to home and professional cooks as well as support applications such as dietary management and cultural preservation. The developed system will be assessed using both subjective (by evaluators) and objective (via the BLEU and ROUGE scores) criteria, to be sure the system works and produces relevant content. The work can be related to the further development of the field of AI in the area of culinary, since it led to the obtained result, a flexible convenient and intuitive tool that can make food more enjoyable.

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Abbreviations

AI – Artificial Intelligence

BLUE – Bilingual Evaluation Understudy

CNNs – Convolutional Neural Networks CV

CV – Computer Vision

iOS – iPhone Operating System

NLP – Natural Language Processing

RAG – Retrieval Augmented Generation

RNN – Recurrent Neural Network

ROUGE – Recall-Oriented Understudy for Gisting Evaluation

UI – User Interface

CHAPTER ONE

INTRODUCTION

1.1 Background of study

Food is an important element in human survival, providing energy and serving a great function in culture and identification (Pollan, 2018). The preparation, consumption, and even discussion of food is an inseparable part of everyday life, and that is what the saying, “we are what we eat” denotes (Pollan 2018). In the modern fast paced world, there are many people who look for quick and inventive ways of making attractive meals. The popularity of food blogs, websites, cookbooks is the evidence of the never-ending search for new and interesting recipes (Chen & Jain, 2020). Digital and internet-based platforms have enhanced vast access to resources on cooking for a wide population.

There is a significant impact of the expansion of online food databases as well as the automation of cooking-related work (Lee et al., 2020; Srivastava et al. (2024); Chhikara et al. (2023). This explosion of digital food has in return increased studies on automation of food recipe generation from food images. Instead of a recipe person has to cook, this technology performs analysis of food images and provides detailed recipes that are extremely helpful for the vast array of users (Srivastava et al., 2024). The process is achieved through the use of computer vision (CV) and natural language processing (NLP) techniques used to interpret the visual aspect of food and modify it to a recipe in the form of text (Chhikara et al., 2023). The main issue is to overcome the semantic gap between visual food data and symbolic representation of recipes (Pham et al. 2021 and Raboy-McGowan et al. 2023).

Lately, there has been progress in the field of artificial intelligence (AI), especially deep learning which has indicated promising progress in overcoming some of the core difficulties when it

involves formulation of recipes (Zhu et al., 2021). Deep learning became a common approach to solving some of the issues which are almost fundamental in recipe formulation. Cooking is usually viewed as an art mixing the taste and ingredient to come up with a masterpiece. This project will seek to explore development of a recipe generator that uses deep learning in improving computational creativity in food preparations. The idea comprises the development of the system that mimics human creativity within which new recipes can be developed from food images. The length of instructions and the number of instructions for each dish as well as numerous food items in one food image greatly pose a challenge (Srinivasamoorthy & Savant, 2022). The system seeks to circumvent these barriers to include the generation of compressed embedding of culinary instructions which are highly similar to the given instructions.

The Recipe Generator is a project that relies for the most part on AI, and it would change the passing on of the recipes and cooking (Gupta et al., 2019). Deep learning and natural language processing (NLP) mechanisms built into the system are able to support personalized recipe suggestions and make the cooking process more efficient. This form of approach has the potential of transforming the way people relate with food. A perfect system would allow users to upload a food photo and get back immediately with the detailed recipe that would help amateur and professional cooks and maybe contribute to a better understanding of the relation between food presentation and the culinary skills. Beyond personal use, such systems have potentials for use in dietary management, personal meal planning, even in preserving culture through digitization and sharing of traditional recipes.

The creation of successful image-to-recipe generation systems implies the resolution of a number of scientific challenges. The food images' appearance can differ significantly because of illumination differences, camera point of views, and methods of its presentation. It is a complex endeavor to analyze the picture of the dish and identify their components to understand its

components. It is also a challenging task to organize coherent instructions via natural language processing (Wang et al., 2022; Ujwala et al., 2023). Moreover, broad access to large databases of high-quality data is required for the implementation of deep learning algorithms, and it is challenging to get it in a consistent manner. The evaluation of generated recipe quality is still a challenging task since the traditional automated metrics do not always coincide with human evaluation (Lee et al., 2020).

This project aims at finding a solution to these difficulties by constructing a strong and precise image-to-recipe generation system. The system will make use of a huge dataset of food images with associated recipes to train deep-learning models that would be able to extract relevant visual features as well as generate textual recipes with high accuracy. It is believed that the obtained research results will help move forward the AI in the domain of cooking and open new opportunities for innovative uses that will make the food experience more appealing for the users.

1.2 Problem Statement

Traditional approaches to recipes retrieval are mainly text-based, and require a known dish name, or ingredient, for search. This reliance on keyword searches can be a significant stumbling block for many of the users, especially those who are new to some of the food styles, or who have witnessed a particular dish, but cannot call it to mind, much less create it. This constraint makes culinary experience and access to diverse food cultures hard (Chen & Jain 2020). The current systems of food recognition as well cannot provide accurate recipe recommendations of food identified due to such problems as: lack of adequate dataset, occlusions, variation on food presentation, and intra class variability. Variations of angle, lighting and plating configuration further increase the complexities of automated identification of food ingredients (AbdElrahman & Yuan, 2020). Furthermore, the current systems lack the flexibility for personal dietary restrictions

(e.g. vegetarian, vegan, or gluten-free diet) restricting a user's diet further and limiting the options for the user even more, making such profiles more relevant. There are some AI based solutions but they typically don't have a handy mobile UI or fail to move efforts to speed, intuitiveness, and availability all of which are fundamental characteristics of successful interaction in mobile platforms. To address such constraints, this project has opted to produce an intelligent food recipe system using the deep learning techniques. The system will give users possibilities to receive cooking instruction and food ingredients on frequency of food images that will make this cooking activity more intuitive and accessible. This innovation will enliven the sense of gastronomic inventiveness and healthier meals, and unite results of food visualization research and practical knowledge on recipes.

1.3 Objectives

1.3.1 General Objective

To create an AI-based system, which is capable of producing accurate cooking instruction, ingredient lists, and a nutrition-facts information directly from an image on food which is going to be used by Eusbett hotel.

1.3.2 Specific Objectives

1. To use deep learning methods for examining food images in order to find main components, amounts of ingredient, and cooking techniques.
2. To include Natural Language Processing (NLP) to bring the identified ingredients and methods to structured recipe models.
3. To develop an easy-to-use mobile app that would enable users to easily come up with recipes using food pictures hence improving the cooking experience.

1.4 Significance of the Work

The purpose of this project is to provide a solution to the real-world problem of effectively and productively creating cooking recipes from pictures of food. The offered methods of recipe discovery tend to require manual research of either cookbooks or Internet sources, which might be very time- and effort-consuming and do not necessarily display personalized preferences and the visual manner to present a dish.

This problem can be resolved by this project as it helps to automate recipe generation process through deep learning methods. With the ability to allow the user to add a food image and retrieve the appropriate recipe, the system allows for a simplified user experience with easier discovery of recipes.

The techniques used in the current project are novel and favorable in various ways. First, the application of deep learning models such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) or Transformer, allows the system to take learnings from complex relationships between the visual food data and textual recipe information (Zhu et al., 2021). This technique allows developing more precise and detailed recipes than the traditional methods do. Second, the system overcomes the problem of the semantic gap between the text and the images, which is also the major obstacle for cross-modal data processing (Pham et al., 2021; RaboyMcGowan et al., 2023).

The potential social implication of this project is great. It can revolutionize people's interaction with food including empowering home cooks to invent new dishes, support professional chefs' culinary products innovation, and perhaps support people to understand the interrelation between food presentation and culinary techniques (Gupta et al., 2019). Furthermore, the system can be

customized to be used in the diet management, a personalized meal plan, and the cultural preservation using traditional recipes digitization and sharing.

The individual goals of the project help to make it meaningful in determining the areas of critical issues of image-to-recipe generation and their mitigation. By the development of a tool that is capable of retrieving salient visual features from images of food and then providing fluent and richly textual recipes, the project moves towards the creation of a useful and convenient tool of various applications.

There were a number of gaps in existing systems or methods that are being addressed in this project. Existing recipe generation systems are unable to cope with the variety of the food image appearance and difficulty of transferring visual information into natural language (Wang et al., 2022; Ujwala et al., 2023). Also, there are no systems capable of dealing with nuances of culinary instructions and developing recipes that are precise and creative. These gaps will be closed by this project through the creation of a stronger, more flexible system.

This project could lead to the further research and practical use in several spheres. It may result in the increase of advanced AI culinary tools like real-time virtual assistants who would give guidance based on food pictures. It would also be helpful in the fields of food science as it would help understand the correlation that exists between the appearance of food, taste, and their nutrition content. Moreover, the developed techniques might be employed in other domains that use the cross-modal data processing, such as image captioning, video description, and product recommendation.

1.5 Scope of the Project

The primary purpose of this project is to create a system that can automatically develop recipes for cooking food based on the images of foods, pursuing the goal of developing the tool that supports

culinary creativity through the more effective and user-friendly methods of discovering recipes. The particular field of research is the implementation of artificial intelligence (AI) namely deep learning and computer vision to the sphere of food and recipe generation, with an emphasis on the intersection between image analysis and natural language processing (NLP) to overcome the discrepancy between the visual presentation of food and words describing recipes (Pham et al., 2021). Raboy-McGowan et al., 2023). Technologies and methods currently used are convolutional neural networks (CNNs) for deriving features from images of food (Zhu et al., 2021), recurrent neural networks (RNNs) or Transformer for deriving the recipe text and a large repository of food images and recipes such as Recipe1M+ (Marin et al., 2018).

Performance of the system will be measured via an amalgamation of objective and subjective metrics; where objective metrics will have BLEU and ROUGE scores for measuring the similarity between generated and Ground truth recipes and Recall@K as the measure of ability of the system in retrieving relevant recipes (Lee et al., 2020). Subjective evaluation will require The end product or deliverable of the project is a working software system that will be able to utilize an image of a food item as an input and give it its corresponding cooking recipe as output. this system will consist of the user interface for the image upload and recipe output, the deep learning models and algorithms. The anticipated scenario and implementation setting for the system is a web or mobile software that would be accessible to home cooks, professional cooks, and any person interested in food and cooking, and it could also be an addition to other applications, such as software for managing recipes or planning diets.

The particular limits and restrictions of the project include the system functioning based on the quality and diversity of the training data, the system might struggle to produce accurate recipes for highly complex and/or ambiguous food images, and that the system may be unable to automate the

variations of the cooking styles and cultural cuisines. Implementation plan or rollout strategy has step-by-step implementation process like data collection and preprocessing, deep learning models development and training, system integration and testing, user interface designing and development and deployment and evaluation. The features to include in order to further enable usability and interpretation include the user-friendly interface of allowing for entry of images and the display of recipes, clarity and conciseness in presenting ingredients in recipes with instructions contained therein, providing for the saving and sharing of recipes by users, and mechanisms for feedback from users to improve the system's performance. This project helps future development and research to advance state-of-the-art in the field of image-to-recipe generation, develop an area for further studies in the use of AI in the culinary field, inspire development of new tools and applications for improvement of food experience, and overcome hurdles in cross-modal data processing and machine learning.

1.6 Summary of the Method

For this project, food images and recipes for the pictures will be the main source of data (Marin et al., 2018). These data shall undergo a number of preprocessing processes to make them suitable for the deep learning model. Image preprocessing will include resizing the images to expose all the characteristics of the images at a standard resolution and normalizing pixel values. Such techniques as rotation, flipping and cropping can be used to expand the data and enhance sturdiness of the model (Zhu et al., 2021). Recipe text will be standardized, tokenized and even deprived of stop words in order to make it ready for natural language processing.

The fundamental part of the project will be related to the use of deep learning models. The food picture will be utilizing CNNs to obtain relevant characteristics using crucial visual features (Lee et al., 2020). Recurrent Neural Networks (RNNs) or Transformer models will be used to create the

recipe text along with ingredients and instructions out of the captured features from the image (Pham, et al., 2021; Raboy-McGowan et al., 2023).

The models will be trained on the preprocessed data with the performance being validated by a held-out test set. The measures to be used in the evaluation will include BLEU and ROUGE scores in order to measure similarity between generated and the reality recipes. Subjective evaluation by human judges will also be used in order to evaluate the quality, relevance and coherence of the generated recipes.

The trained model will be implemented into a software program, and the model will have userfriendly interface whereby the image input and recipes will be entered. With this application, the users will be allowed to upload a food image and get a recipe generated.

After development, the system will be deeply tested in order to check whether it is functional and accurate for use. Such testing process will include assessing how well the system will deliver accurate and coherent recipes for different images of food.

Finally, in the system, there would be a user feedback mechanism through which the users would be able to add inputs regarding the produced recipes. This feedback will be incorporated in the continuous improvement of the models towards better performance of the system in future.

CHAPTER TWO:

LITERATURE REVIEW

2.1 Introduction

Automatic generation of cooking recipes from pictures of food has developed as the promising field of research because of the increased access to online food content and the need to automate cookery-related tasks (Lee et al, 2020; Srivastava & al., 2024; Chhikara & al., 2023). This technology aspires to analyze the food pictures and provide detailed recipes which can help numerous users (Srivastava et al., 2024). The process encompasses the use of techniques of computer vision (CV) and natural language processing (NLP) for retrieving visual characteristics of food and their rendering in the form of textual recipe (Chhikara et al., 2023). The major issue is to close the semantic gap between visual food data and symbolic representation of recipes (Pham et al., 2021; Raboy-McGowan et al., 2023).

There is tremendous potential for using image-to-recipe generation, which is likely to transform how people interact with food. The ideal system would allow the user to take a food image and instantly get a detailed recipe that will help a homemaker, a professional chef, as well as contribute to the better understanding of the connection of the food presentation and culinary practices. Beyond personal applications, the systems could be applied in any dietary management program, customizing the delivery of meals and in some cases preserving cultures as traditions are turned into modernized formats through the digitalization of time-tested recipes.

Yet, for the creation of feasible systems for image-to-recipe generation, one must address a number of scientific challenges. The presentation of food images might be extremely different depending on the variability of illumination, perspectives of cameras and methods of presentation. Analyzing

the image of the dish in order to understand what elements compose is a challenging activity. It is also a difficult challenge to make coherent instructions with natural language processing (Wang et al., 2022; Ujwala et al., 2023). Moreover, even to apply a deep learning algorithm, it is necessary to get an access to databases that are overloaded with the best-quality data, and it is not always easy to have these sources. The measure of generated recipe quality is still a difficult task, as those conducted automatically may not be consistent with human evaluation most of the time (Lee et al., 2020).

This literature review digs into the progress in AI and deep learning methods involved in food images analysis, and recipe generation considering the flaws of existing systems, and it reveals possibilities for further studies. It explores the main ideas, details the systems and methods that are used, and reviews the constraints and the difficulties in this emerging field.

2.2 Definition of Concepts

Several core technical terms are central to this discussion.

2.2.1 Convolutional Neural Networks (CNNs):

CNNs are a group of deep learning models that have shown tremendous performance in image analysis problems. They are automatic and learn hierarchical representations of visual data so that they can extract image features (He et al., 2018). The CNNs have emerged as a staple in the process of computer vision, proving effective in a wide range of tasks such as image classification, object detection, or image segmentation.

- How CNNs work: A CNN has several layers, namely, convolutional layer, pooling layer, and a fully connected layer.

- Convolutional layers are the bricklayer of a CNN. They use filters (or kernels) to the input image for detecting local patterns like the edges, textures, and shapes. The size of the filter will determine the patterns' spatiotemporal extent that the filter can detect. Each filter is swept across the picture and at each point a dot product is performed between the filter and the pixel material in the region of the image. As an outcome of this operation, the feature map is a result which indicates how the filter has responded in the image. Several filters are usually implemented in each convolutional layer to extract different elements of features. The convolutional step may be made as:

$$G[m, n] = (f * h)[m, n] = \sum_j \sum_k h[j, k] f[m - j, n - k]$$

- The input image is f, kernel is g, output/feature map value at location (x,y) is (f*g) (x,y).
- Pooling layers are the ones that are implemented in order to reduce the spatial dimension of the feature maps. This helps reduce the parameters in the network hence becoming more computationally efficient and less prone to overfitting. Pooling also contributes to make the model robust to small changes in input image. Examples of some of the common pooling operations are average pooling and max pooling. In pooling, the max pooling uses the maximum of a small area in a rectangular-shaped format whereas the average uses the average.
- The fully connected layers tend to be addressed on the last end of the network. These merge the high-level features learned by the convolutional and pooling layers in order to come up with a single final prediction. A fully connected layer has the neurons hooked up to those of the previous layer. The last classification or regression can be attained on fully connected layers.

- Such a hierarchical feature extraction process makes CNNs very efficient for such recognition tasks as, for example, food image recognition and they can learn to recognize the ingredients and particular visuals of various dishes (He et al., 2018). The CNN's capacity of learning more complex features in deeper layers is very important for identification of the differences in food type, which can be very subtle in visual terms.

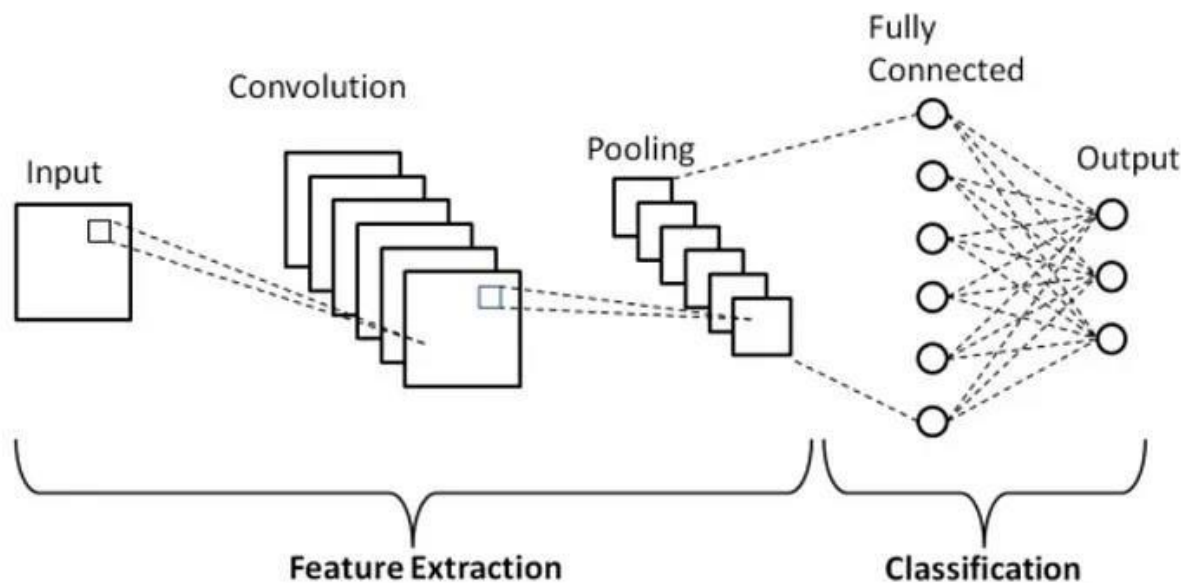


Figure 2. 1 (Convolutional Neural Network (CNN) structure)

2.2.2 RNN (Recurrent Neural Network):

RNNs is a type of neural network that is meant to deal with sequential data, in this case the text. RNNs therefore differ in the structure of the feedback loop compared to feedforward neural networks enabling the former to have memory of past input values. This memory allows them to extract dependencies between elements of a sequence, which is vital for such tasks, for example, generating a coherent sentence or correctly extracting the context of the text.

- How it works: they work, as each element is considered while preserving a “memory” for the previous elements in a “hidden state”. This latent state is renewed on every new word

of the sequence, which allows the network to consider the context of this sequence. In every turn, the RNN can make a prediction at the next element (such as the next word) conditioned with its current hidden state. The training of the network prepares it to predict such patterns by properly adjusting its internal weights and the parameters that minimize the error of prediction between output and actual elements in the sequence.

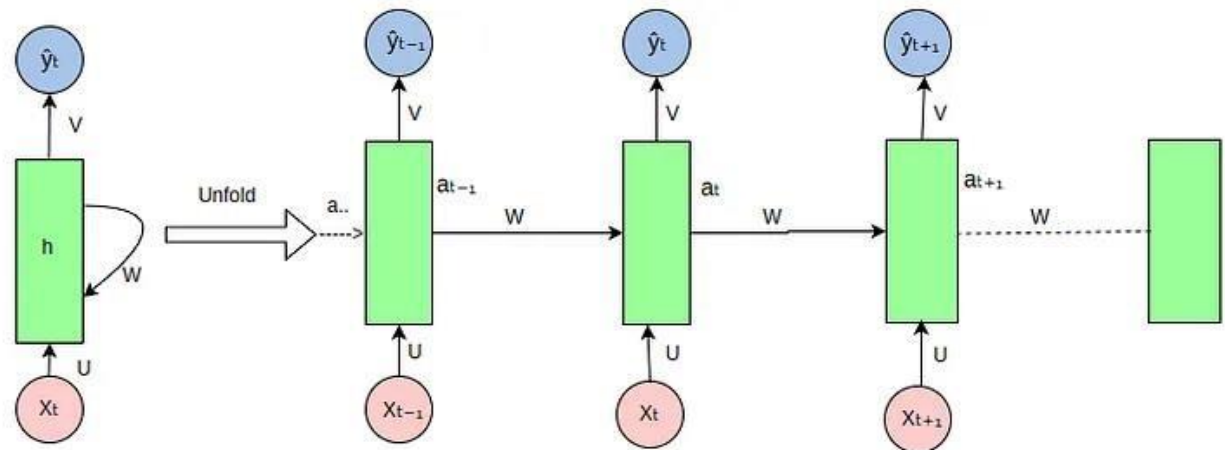


Figure 2. 2 (Recurrent Neural Network (CNN) architecture)

2.2.3 Long Short-Term Memory (LSTM):

A family of RNN architectures which overcome the vanishing gradient problem, hence; the model learnt long-range dependencies in sequential data (Hochreiter & Schmidhuber, 1997). The vanishing gradient problem is a known problem with training deep neural networks, because as gradient are back propagated, they can get very small through many layers of the neural network that will make it hard for the network to learn long-range dependencies. This approach is avoided by LSTMs, which propose a gating system that involves that the flow of information from entering through the network.

- How it works: LSTMs add “cell state” (long-term memory) and “gates” (forget, input, output), that regulate the flow of information. These gates either add or subtract the

information on the cell state selectively; this way, the network is able to retain relevant information through long sequences. Significantly, the mechanism of using gradients backpropagation over time in LSTMS, where cell state is used, ensures that the network can learn to regulate the flow of the gradient reducing the issues with vanishing gradient and acquiring capability of learning long-term patterns.

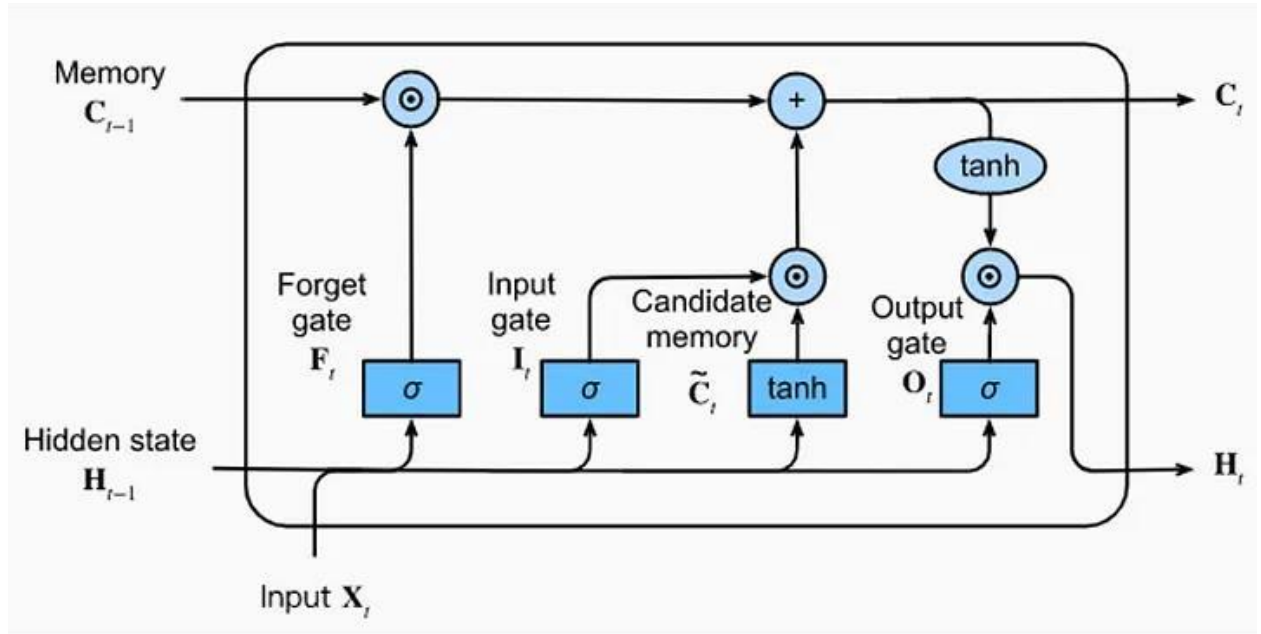


Figure 2. 3 (Long Short-Term Memory (LSTM) architecture)

2.2.4 Transformers:

A more modern architecture that uses attention mechanisms as opposed to recurrence (Vaswani et al., 2019). In contrast to RNNs (which sequentially process pieces of data in one step at a time), Transformers are capable of parallel processing of an entire input sequence. This allows them the ability to capture long range dependency better and more efficiently. Through the attention mechanisms, the model is able to contribute to the significance of the bits in the input sequence in the modeling of a specific element.

Transformers have been gaining a lot of popularity in NLP, attaining the state-of-the-art results in many tasks, from machine translation, text summarization, to question answering.

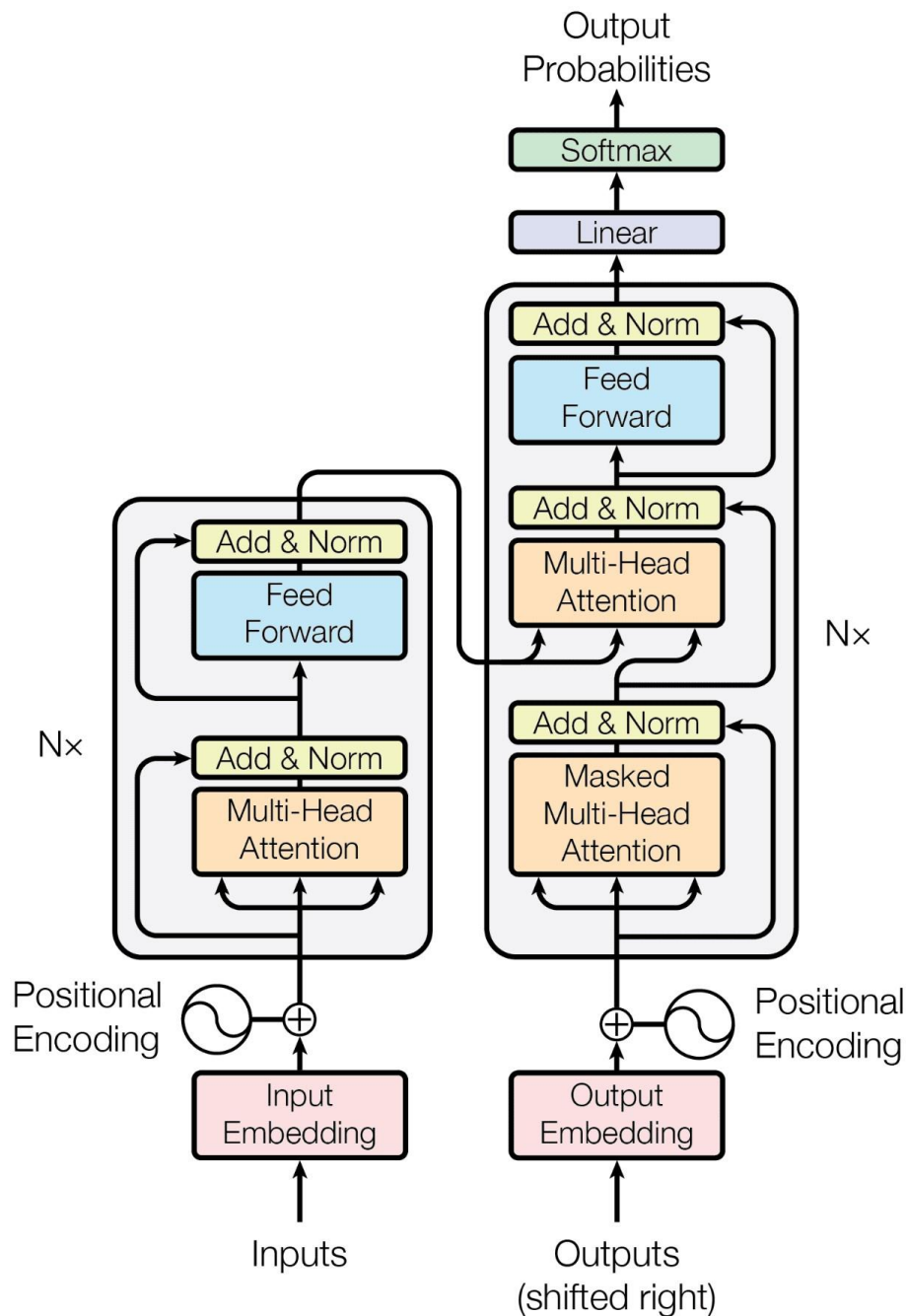


Figure 2. 4 (Transformers architecture)

2.2.5 Cross-Modal Embedding:

This is a technique of mapping data from various modalities (e.g., images-text) into a common vector space. In the frame of the image-to-recipe generation, the use of cross-modal embedding enables the system to encode the images with foods and recipes in such a way that their semantic relation can be reflected (Marin et al., 2018). The objective is to learn a mapping such that embedded representations of semantically similar items (e.g., a picture of pizza along with the pizza recipe) should be compacted in the embedding space, whereas the dissimilar items should be far apart.

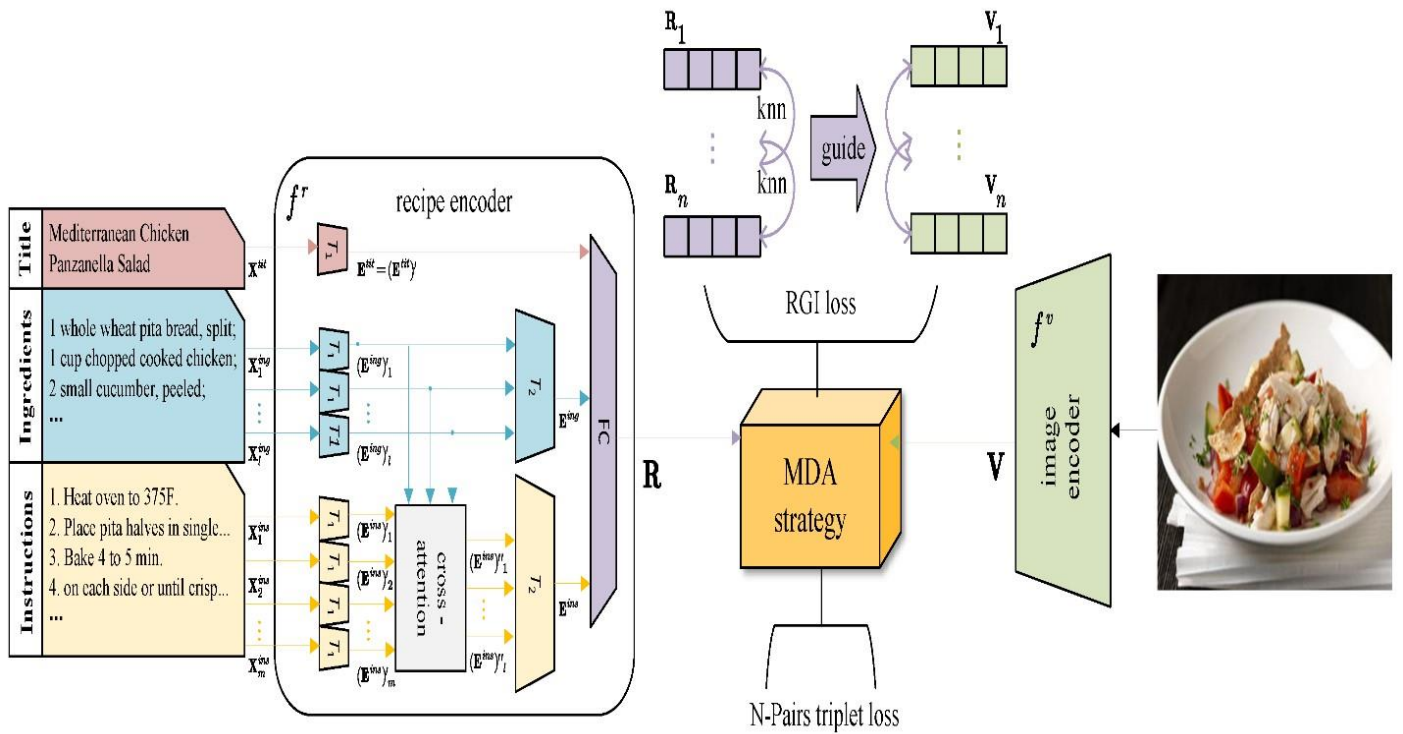


Figure 2. 5 (Cross-Modal Embedding architecture)

2.2.6 Semantic Gap:

The semantic gap refers to the difference in the level of representation between mankind's way of understanding data (e.g., high-level concept such as "pizza") and computers represented as an array of pixel values. This gap is one of the most serious challenges of multimedia analysis such as an image-to-recipe generation. It is easy for a human to relate a food image to the recipe attached because he or she can comprehend what ingredients are used, what steps must be taken in cooking and how the culinary concept was achieved. Yet, for a computer, an image is just a set of pixels and recipe is a succession of words. The issue is to bring about models that are able to learn and be able to associate the various representations and understand the semantics behind the relations.

2.3 Description of Systems and Methods

Automating the process of creating cooking recipes regarding food images is one of the areas, where artificial intelligence (AI) is of utmost importance within this project. It is considered that DL models, particularly CNNs, are frequently utilized for food image analysis because they possess capability to automatically learn hierarchical description of visual data (Lee et al., 2020; Zhu et al., 2021).

2.3.1 Role of Deep Learning:

This task can be done more effectively using deep learning compared to the traditional machine learning because it can be used to handle the complexity and variability of food images where relevant features are extracted without the need for manual feature engineering. Conventional machine learning algorithms often involve hand-engineered features and can be time-consuming, and not provide the intricacies associated with visual information. Deep learning models, in turn, can learn these features directly from the data hence better performance.

2.3.2 Challenges in Food Image Analysis

Examining food images involves a variety of distinct issues that render the undertaking of creating recipes rather challenging.

- **Intra-class variability:** Among the major problems is the wide disparity of the presentation of the same meal (Wang et al., 2022). Variations between a single recipe, like "spaghetti and meatballs," can be made and serve in very different ways. Factors like lighting, angles of the camera, and ways of presenting a dish can do a great deal in changing how a dish is viewed in a picture. For example, natural lighting may make the sauce seen brighter while fluorescent lighting may provide it with a different color. On the same note, close-up shots of images may bring out the texture of the meat balls, while a wide-angle shot will be focusing on general table setting. In addition to this, ingredient differences, as in, using various kinds of tomatoes or garnish, can also cause this variability.
- **Inter-class similarity:** Another problem is the fact that various dishes may have similar visual properties (Ujwala et al., 2023). For instance, it could become challenging to see the difference between "spaghetti with tomato sauce" and "pasta with marinara sauce", especially when taking the pictures under the same conditions. Both dishes are pasta of varying sorts but with the topping of red sauce. Making fine distinctions between ingredients or preparation processes from what one can tell from an image is a tall order.
- **Occlusion and clutter:** At real-life applications, food items are usually partially occluded or located in an environment with other objects (Ma et al., 2024). A plate of food may be adorned with a number of herbs or spices which may hide the main ingredients. Besides, the presence of cutlery, napkins or other table settings may make an image cluttered hence making it hard for a system to focus and separate out the relevant food components. The

model has to be strong enough to support these insights and distractions if it is to make an accurate interpretation of the food image.

- **Variation in Image Quality:** Pictures of all kinds can be extremely different in quality because of such reasons as the device for image capturing, the mastery of a photographer, and the existence of the digital artifacts. Photos that have been captured in professional settings through high-resolution cameras will be very different from those captured using old mobile phones.

2.3.3 Existing Approaches

Scholars have studied different techniques of addressing the problem of recipe generation from food images. Among the major aspects of a wide range of these systems is the application of Convolutional Neural Networks (CNNs) to derive meaningful representations from the images (Lee et al., 2020; Zhu et al., 2021). CNN acts as the encoder and processes the input image into a fixed-length vector representing the essence of visuals of the dish. This vector representation is intended to capture the information on the ingredients, the style of cooking and general look of the food.

Then this encoded image representation is passed on to a decoder (usually a Recurrent Neural Network (RNN) or a Transformer model) to produce the recipe text. The RNN or Transformer runs the vector sequentially, thus creating a list of ingredients and cooking instructions of the recipe that is represented by the sequence of words (Pham et al., 2021; Raboy-McGowan et al., 2023). Such decoding process not only demands a model to comprehend what is seen in the image but also requires having information about culinary concepts and language.

A major challenge in this process would be to make sure that the generated recipe is well aligned with the imagery information in the picture. To resolve this, attention mechanism has been used

(Gupta et al., 2019). The decoder can focus selectively on those parts of the image, which contribute most to the description given in every word of the recipe, because of attention mechanisms. For instance, in producing the ingredient “tomato”, the model can focus on the red areas of the picture. This is helpful for the model as it will help it ground the generated text in the visual evidence to enhance the relevance and accuracy of the recipe.

Some of the approaches have taken the step of ingredient prediction as being intermediate (Salvador et al., 2019; Salvador et al., 2019). Rather than directly produce the recipe, the model estimates the ingredients that are in the dish first. This ingredient prediction can then help to determine the subsequent recipe generation process. Specifying the ingredients explicitly, the model can restrict the output space and come up with recipes that are more likely accurate and coherent.

A second avenue of research has examined ways of learning multi-modal embedding of two things i.e., images and recipes (Marin et al., 2018). This is through training a model to capture both images and recipes as vectors in a single embedding space. In this space, kindred images and recipes are near each other while dissimilar ones are far from each other. Using multi-modal embeddings, the retrieval can also be enhanced from images and vice versa regarding the recipes. For instance, with an image of a food, from a pool of recipes, the system would be able to find the most similar by locating recipes with the closest embeddings.

Such image-to-recipe generation systems are usually trained and validated on this large collection of food images and recipes. Among the widely adopted datasets, there are Recipe1M+ (Marin et al., 2018), which includes more than one million cooking recipes and almost thirteen million food images. This dataset is a useful repository in training and evaluating deep learning models for this

task. The prevalence and variety of Recipe1M+ give models an opportunity to extract complex relationships between visual food appearance and monetary recipes.

The need for productive and enterprising methods of discovering and preparing meals has gathered momentum and this is what propels the formulation of these systems. This demand has been driven by people's busy lifestyles, as well as the increased popularity of food-related content on social media (Chen & Jain, 2020). Platforms such as Instagram, where people constantly post pictures of the food that they eat have revolutionized the way people relate to culinary information (Ujwala et al., 2023). This has brought us wealth in visual food data that has the potential to be used to train image-to-recipe generation systems.

But there are issues relating to the use of social media images as well. Food photos posted on such sites as Instagram often have a great difference in quality, appearance, and style. Some of the elements that can significantly influence the appearance of food on the screen include the lighting conditions, camera angles, filters, and editing techniques. This inconsistency makes it challenging for image-to-recipe generation systems to comprehensively get a clear picture of the content of the images and deliver reliable recipes.

Although there has been a lot of advancements in this field, but there are still huge gaps, when it comes to accurate and creative transference of food images into usable recipes by recipe generators.

2.4 Identification of Gaps

Although, AI has made significant progress, technical issues that restrict the accuracy of current systems still exist.

2.4.1 Challenges and Limitations:

These involve issues with dealing with the differences in the appearance of food images including changes in the images' illumination, the angle of capturing and food presentation (Ma et al., 2024; Wang et al., 2022). Such variations may get in the way for models in identifying the ingredients and cooking methods alluded in the picture.

Creating guidelines for cooking in detail and accurately is still a major challenge as well (Wang et al., 2022; Ujwala et al., 2023). Here, besides the ability to understand the visual part of the image, the model must have knowledge of cooking procedures and cooking techniques.

Hallucinations: Sometimes, large language models (LLMs) employed in recipe generation generate recipe steps and or ingredients which are not in reality in the image, reported Liu et al. (2024).

- **Lack of precise ingredient measurement:** Existing models are not always able to give the correct quantity of ingredient or measurements.
- **Difficulty with complex dishes:** The creation of recipes for complicated dishes with numerous ingredients and complicated procedures of preparation is still the task.

2.4.2 Data Limitations:

The quality of the deep learning models highly depends on the availability and quality of training data. Although such datasets, as Recipe1M+, can be really massive, they probably could not involve the entire variety of food forms and the diversity of cooking styles existing in the real world.

Data bias might also be a problem because data sets may be biased toward some cuisines or cooking techniques and so the models do not generalize to different types of food.

The lacunae of the present systems' accuracy and reliability can influence the decision-making regarding food, which might impede their utility in practical circumstances.

2.4.3 Impact of Gaps:

When a system brings about inaccurate recipes or recipes that do not have all the required ingredients, it may result in wrong cooking, and it might even contribute to waste of food.

In the case of Diet planning or personal diet plan, inaccurate recipes might have more serious results.

These gaps give the need for more research to build stronger, more accurate, and user-friendly image to recipes generation systems. It will be essential to overcome these limitations for mass implementation of this technology and adoption of its use in different areas.

CHAPTER THREE:

METHODOLOGY

3.1 Introduction

In this section, the method of developing the food image to recipe generation system is described. The section covers the study approaches, steps involved in data collection and refinement, the algorithm coding process, system building tools, system organization, and the visualization of data. The Agile methodology forms the basis of software development, with the emphasis being put on iterative advancements and collaboration, flexibility.

3.2 Type of Research

The applied method adopted by the study is intended to solve the practical problem of automatic generation of recipes from food images. Computer vision is leveraged in this effort, along with deep learning techniques, and natural language processing (NLP) to engineer such a system which can be deployed. There is also an exploratory side of the research, which was created to analyze how a wide range of deep learning models – different CNNs and RNN/Transformer architectures – can be used to fine-tune the performance of the system.

3.3 Agile Development Methodology

Agile is chosen as the development approach because it is step by step, matching the project's focus on mechanism updates, feedback integration and flexible phases. Using this approach, teams can update and modify the system from cycle to cycle through steps like preparing data, modeling and building a user interface (Jackson et al, 2019).

Alternatively, applying the Waterfall idea to a project scenario could illustrate that its strict and phase-driven way (requirements, design, implementation, testing, deployment) would be best suited to projects where the scope is very set and needs for changes are slim (Adolph, 2018). Within such a system, if recipes are taken directly from databases and created without updating the model, the Waterfall model could guarantee a steady progression. Notably, owing to the fast evolution of AI optimization and the continued need for user input, Agile is the best approach, according to newly published research (Jackson et al., 2019). It is especially noticeable how Agile works best when bringing together practical and technical factors in culinary AI systems (Kokol, 2022).

Instead of one big project, Agile spilt into small, manageable sections called sprints. Every of these iterations is created with the goal of bringing a few key pieces of the project to life. An example would be that in one sprint, the function to put in and process food images is built, then the next sprint could focus on bettering the model for telling ingredients apart and after that, focus on displaying completed recipes through the user interface. Thanks to structured development, the system grows step by step and every upgrade adds value to the total project goal (Gemino et al., 2021).

The stand-up is a basic meeting that happens daily in Agile. Each short session allows team members to inform others of their progress; explain any problems they encounter and agree on the main goals for the day. For a food recipe generation project, having daily stand-ups encourages every team member to stay on key tasks and act fast as the product is developed (Balland et al., 2018).

All of the features, improvements and bug fixes planned for the project are part of the Agile backlog. The priority is always to move tasks from the backlog that most satisfy the user's top needs. In the beginning, it is important to ace the system's performance, like being able to correctly

generate recipes using various food images. Following sprints would concentrate on upgrading the user interface to make generated recipes available and manageable by cooks and chefs working in their own homes or kitchens. Because such an approach makes development flexible, crucial features are built at the beginning and the project can be improved as users and testing reveal what needs to be modified throughout the process (Serrador & Pinto, 2019).

Moreover, Agile relies heavily on testing and blending together work at regular intervals. It matters a lot in AI because often we need to verify model accuracy and performance all the time. The model components built or updated within each sprint may be extensively tested on some of the data and their findings should measure up to known guidelines. As testing is repeated, the system keeps getting better until it finally reaches a level of accuracy and reliability acceptable for endusers (VersionOne, 2019).

Especially, all feedback from stakeholders is regularly used during all stages of developing the project. Always consulting experienced cooks, chefs or avid home cooks keeps the food recipe project on track with practical recipes and what users look for. Once the first functioning model is completed, chefs and cooks can share feedback about how convenient the display is, if the instructions are suitable and what else might be useful in the kitchen. The approach helps to make the final product both technologically sound and simple to implement when making real meals

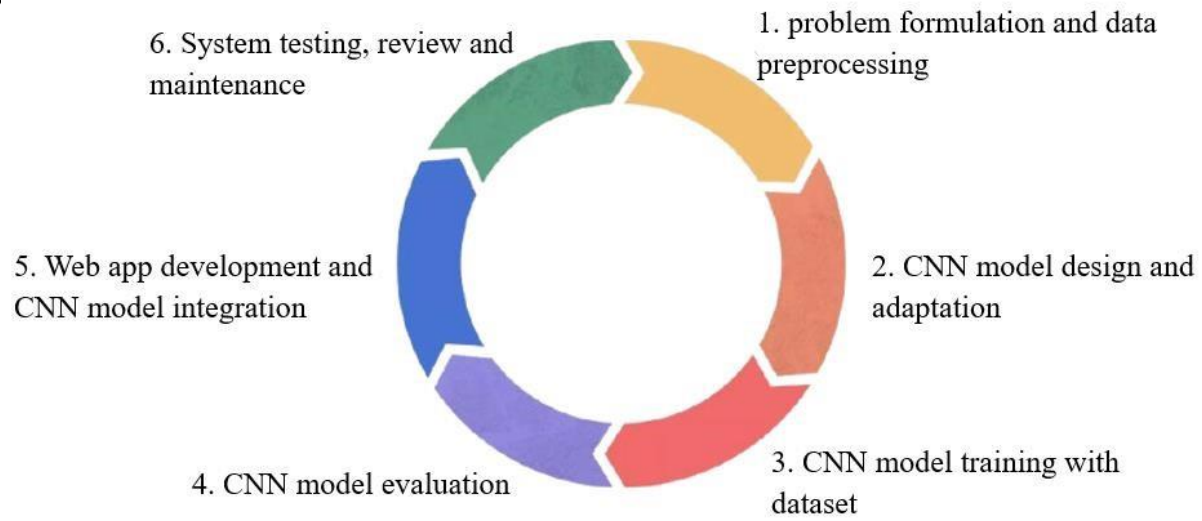


Figure 3. 1 (Agile methodology)

3.4 Data

Data quantity and quality are fundamental to the achievement of the best results by the system. The core dataset behind this study is formed by Recipe1M+, which has a vast collection of over one million recipes and 13 million food images (Marin et al.,2018). Combining its variety and strict structure, Recipe1M+ facilitates the development of sophisticated models that can handle both information about a recipe and pictures together.

- Image Data: Food pictures from Recipe1M+ demonstrate many dishes in visual representations, which underline differences regarding appearance, ingredients, and presentation.
- Recipe Data: The recipes of the dataset are accompanied by lists of ingredients and indepth instructions of cooking, connecting textual description with visual ones.

3.4.1 Data Preprocessing

Systematic processing of the data with the help of several preprocessing techniques provides compatibility of the deep learning models.

- **Image Preprocessing:** All images are converted to an identical resolution (e.g. 256x256 pixels) in order to retain images of an equal size for CNN inputs respond. Image transformations such as rotation, flipping, and cropping are performed in order to increase the training set and improve the performances of the model against variations in food image appearance.
- **Text Preprocessing:** Standardization efforts for the text include cleaning up other characters, correcting the spelling errors and ensuring that the entire document is in lower case.

Before NLP models preprocess the text, it is tokenized into words or sub words to be transformed into numerical vectors for use in the models.

By stop words, they may be dropped, but this step is not necessary for many recipes since simple words such as “and, then, until” may contribute valuable meaning. This ensures important baking directions are protected even in their use of simple, meaningful words such as, “then” or “until”.

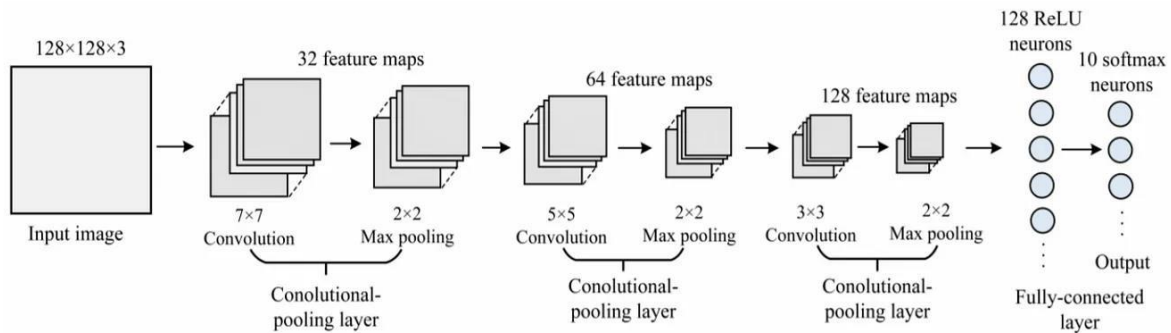


Figure 3. 2 (Data Preprocessing architecture)

3.5 Algorithms

A number of algorithms of the areas of computer vision and natural language processing are used in the system;

3.5.1 Convolutional Neural Networks (CNNs):

CNNs are regarded as important deep learning models because they show great results when working on image analysis. They use this method to automatically build a structured system for analyzing images and picking out the most important parts (He et al., 2018). Most of the image analysis in the project relies on CNNs.

Raw food images go through their system, allowing them to spot key highlights and match unknown ingredients (Srivastava et al., 2024; Ma et al., 2024; Raboy-McGowan et al., 2023). Many of these layers work together. Convolutional ones scan for edges and textures, while pooling layers shrink the image to make it more robust. Next, fully connected layers use all the high-level information to make their prediction about the image's content.

Raboy-McGowan et al. use ResNet-50, ResNet-101 and DenseNet-121 as part of this project's CNN architecture (2023). These models are picked because they do a good job at classifying food pictures and describing their features, each model offering different levels of overall performance for the amount of computation needed. Improvements in identifying food in photos are best achieved when using models such as DenseNet-121.

With their ability to recognize small details in images, CNNs can easily tell what food items are, no matter how different the photographs appear. The information from the CNN forms the basis for creating the recipe, as CNN delivers the image's features in the form of a fixed-length vector.

3.5.2 Recurrent Neural Networks (RNN), and Transformers Networks;

RNNs are a type of neural network made for working with data that is presented in a sequence, like text. They read a word by word, recall the past state in a second "hidden" layer and change this state as each next word in the sentence arrives. So, the network can use the context of the sequence to choose what will come next (Pham et al., 2021). With LSTM, a family of RNNs, the issue of disappearing gradients is handled which means the model can discover dependencies in data over long series (Hochreiter & Schmidhuber, 1997).

Recently, sequence-to-sequence tasks have relied on Transformers which apply attention instead of relying on recurring processes (Vaswani et al., 2019). They are effective in spotlighting relationships across many parts of an input by handling all the words at once and assessing how remarkable each one is. With this ability, the instructions given by the system sound more smoothly and clearly (Lee et al., 2020).

RNNs (mostly LSTMs) and Transformers are necessary in the food recipe generator to work with recipe text and provide cooking directions. RNNs had always been used, but today, Transformers are chosen for generating instructions in easy-to-understand and relevant terms. Capturing the complex and long-range links between details in a recipe is more straightforward with the selfattention mechanism in transformers.

3.5.3 Multimodal Embeddings:

There is a gap between food images and textual recipes and so multimodal techniques are needed to link them (Pham et al., 2021; Marin et al., 2018; Salvador et al., 2019). The method involves placing both photos of food and their recipes on the same vector map. Similar images and recipes tend to be together in the common space, while images that are not similar and recipes for dissimilar meals, are far from each other.

How effectively the system links images and texts is driven by how good the embeddings are. Training models to learn joint embeddings for recipes and their related images has been supported by very large datasets, one of which is Recipe1M+.

They matter because they help quantify how much a recipe matches its associated food image. This feature helps the food recipe generator to perform major tasks. As an example, they make it possible to match recipes with an image, making search more effective. Moreover, they lead to new cooking instructions that fit the design of the target image, keeping the recipes realistic. Because multimodal embeddings notice the connections between text and images, the recipes they produce tend to be both cohesive and valuable.

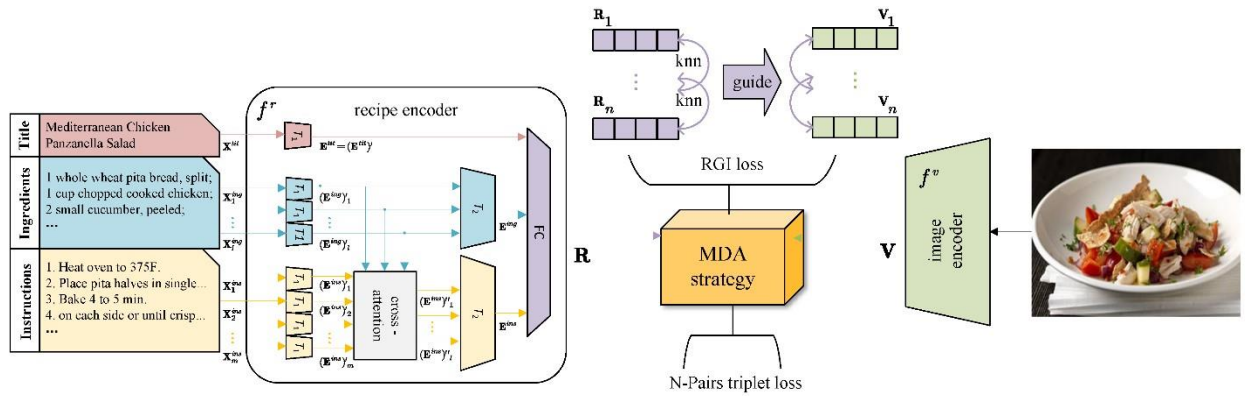


Figure 3. 3 (Multimodal Embeddings)

3.5.4 Retrieval Augmented Generation (RAG):

By including a retrieval process, Retrieval Augmented Generation (RAG) makes content generation more advanced. First, RAG looks up relevant information in a huge database, instead of just depending on learned parameters. The generative model is improved by including facts memorized and used, along with its ability to communicate clearly (Liu et al., 2024).

The main aim of RAG in recipes is to prevent “hallucinations,” where the model produces options that do not make sense (Liu et al., 2024). Because the system looks for related recipes, RAG guarantees that what is produced is based on correct cooking knowledge.

RAG will help the food recipe generator project perform much better. As soon as a user takes a picture of food, the system examines it and detects what the major ingredients are. They become a query for the next part of the process. This part of the system explores a big recipe database such as Recipe1M+ (Marin et al., 2018), to look for recipes that are visually alike or include some of the same ingredients.

The collected recipes are given together with the image attributes to the generative model (for example, Transformer or RNN). Using real-world facts as a base makes the recipes from the model both accurate and believable for cooking. So, if the reference is a pasta dish image, RAG will pull pasta recipes so the writer can create a cohesive text.

3.6 Evaluation Metrics

The models are subject to evaluation by the following set of metrics:

- **BLEU (Bilingual Evaluation Understudy)” and ROUGE... (“Recall-Oriented Understudy for Gisting Evaluation”:** In order to compare the degree of close fit of the output recipe text with the original recipe text. These metrics will compare times of the overlapping n-grams between what the model produces and what is intended in text.

$$\text{ROUGE-N} = \frac{\sum_{S \in \{ReferenceSummaries\}} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in \{ReferenceSummaries\}} \sum_{gram_n \in S} Count(gram_n)} \quad (1)$$

Figure 3.4 (ROUGE-N formula)

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}.$$

Then,

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right).$$

The ranking behavior is more immediately apparent in the log domain,

$$\log BLEU = \min(1 - \frac{r}{c}, 0) + \sum_{n=1}^N w_n \log p_n.$$

In our baseline, we use $N = 4$ and **uniform weights** $w_n = 1/N$.

Figure 3.5 (Bleu formula)

- **Recall@K:** To get the range within which the system retrieves (extracts) relevant recipes from the database. The measure is whether the target recipe is part of the top K returned results of the system.

$$\text{Recall}(R)_k = \frac{|\{r \in R : r \leq k\}|}{|R|}.$$

Figure 3.6 (Recall@K formula)

- **Human evaluation:** In order to identify quality, appropriateness, and coherence of the generated recipes. Experienced judges will give comments to the created recipes with attention to clarity, correctness, and level of quality. We will seek more inputs from users to improve the system. Besides that, statistics related to user experience (UX) and usability scores will be taken into account as well because, due to the nature of the system's user interface, they are important to consider. Respondents will complete a Likert-scale questionnaire while also giving feedback as part of a recipe usability-testing protocol.

3.7 Development Environment

The system's development comes through the combination of different technologies:

3.7.1 Flutter:

The UI of the food recipe generator application is developed mainly with Flutter as the preferred frontend toolkit. Thanks to being developed by Google, Flutter is a strong, open-source kit that lets developers write applications once and use them across mobile, web and desktop environments. The fact that it works on different platforms is a plus which means just one development is needed to provide the application to a wide group of users across devices (Google, 2023).

Flutter is chosen for this project because of several important strengths it offers. The importance of good visuals and responsiveness in applications is extra strong for a simple user experience that suits images and displaying recipes. Thanks to Flutter's widget system, you can create modular and easy-to-use user interfaces using the same reusable pieces of code (Nash, 2020). In addition, the ability to fix code in real time while the app is running has a big impact on the development process. The speed of growth benefits greatly from using Agile, allowing us to focus on continuous development and always collecting user opinions (Martin, 2023). With Flutter, the food image to recipe production system enjoys a powerful and flexible way to create a strong and seamless user interface.

3.7.2 Django:

For the food recipe generator project, developers will use Django, the high-level Python web framework. Surely, strong APIs are needed to seamlessly bring deep learning models into the Flutter frontend. Because Django is easy to develop, secure and simple to maintain, choosing it for server-side logic is logical (Holovaty & Kaplan-Moss, 2019). This also allows the project to make

use of Django's ability to build efficient REST APIs. With these endpoints, I am able to link the UI with the backend for the AI to work on the images captured using Flutter.

Django will handle everything on the backend, working with CNNs for images and Transformers to build recipes. Also, Django allows users to manage the database so that important data such as images, recipes and model parameters, are stored and accessed easily (Vincent, 2018). Because of MVC, the program is organized so that the various parts work more independently and are easier to maintain and improve as the application is scaled. The Django APIs use RESTful programming to make sure the frontend and backend can be built and deployed separately. Because of Django's strong backend infrastructure, user requests are processed, AI models are used and generated recipes are returned fast and smoothly. Thanks to its wide range of features for web development and APIs, Django was chosen as the best solution for the project's integration to the backend.

3.7.3 Python:

Building the artificial intelligence models used in the food recipe generator is mainly done in Python. Because Python offers many libraries and frameworks, along with its clear nature and versatility, it's a top choice for tough machine learning applications (Van Rossum & Drake, 2019). PyTorch or TensorFlow is used for both building and training deep learning models. They include many useful features for arranging, running and making use of neural networks, helping to set up CNNs for extracting features from images and Transformer or RNN models for generating recipes (Paszke et al., 2019; Abadi et al., 2018).

Processing text within recipes is made possible by Natural Language Processing (NLP) for this project. Recipe text is preprocessed with NLP, including standardization and tokenization which is essential for the generative models (RNNs/Transformers) that generate detailed and clear cooking instructions (Jurafsky & Martin, 2021). Python is chosen with PyTorch or TensorFlow due to their

powerful features in number computing and deep learning studies, so model development stays flexible and efficient. When you combine this setup, it guarantees better processing of data, model creation and easy integration with Django for a perfect recipe app.

3.7.4 SQLite for Data Storage and History

SQLite has been chosen, since it is small, flexible and stores data locally as files which makes it right for this project's recipes and user history. SQLite is included directly into the application, so it does not need a dedicated server system as other database management systems do (Owens, 2020). Because of this, it's perfect for building mobile or desktop apps where a full database would be too much trouble.

A variety of factors connected to the project guided the choice to use SQLite. Because the serverless architecture stores everything in one file, deployment and management are much simpler and you can transfer the database anywhere you go (Hipp, 2022). Because Flutter integrates with local devices easily, data can be saved locally in the front part of the app, without the trouble of setting up networks. It is possible to keep generated recipes, user inputs and recent images stored on the user's device, so they can still be used offline and accessed quickly.

Django usually uses PostgreSQL on the server, but for keeping data locally we often use SQLite as a cache and as a record of each user's screen view or commonly used recipes. By doing this, the application can bring up past interactions, making it easier to offer a more interesting experience for each person by using their favorite materials.

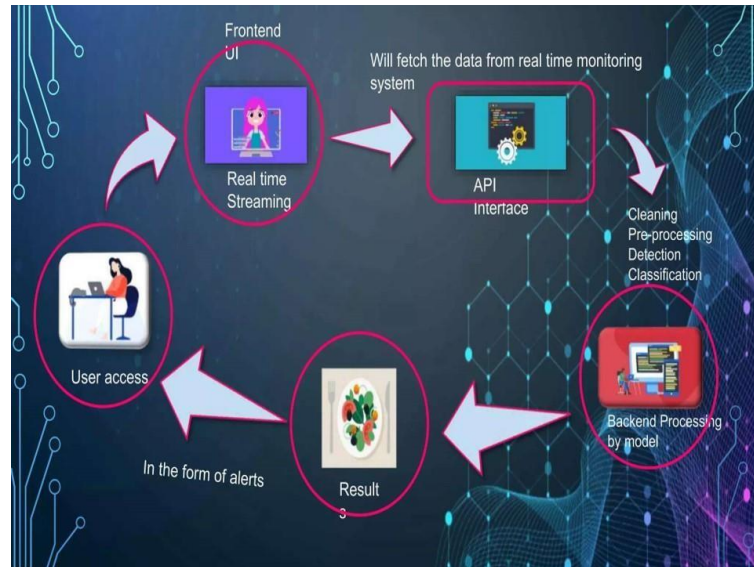


Figure 3. 4 (Development Environment Stack)

3.8 System Architecture

The composition of the architecture of the system is designed to deal with the carrying of data from the user's capture of image to the presentation of the generated recipe. Through the following steps each component in the system became a cohesive part:

- **Image Capture:** Users capture a photo of the food using the Flutter UI. An app with userfriendly interface is presented by the Flutter application – it is possible to either take a new photo or select an existing image from one's device.
- **Image Transmission:** The image file is passed through to the Django backend, accessible to the backend through a server API. Through sending an HTTP request, the captured image is sent to a Flask app by the Flutter application through a particular endpoint on the Django server.
- **Image Processing:** Image analysis is managed by a CNN in Django backend. The Django backend fetches the image imports it and prepares it for use in the CNN model. Image

features that are critical are extracted through CNN processing forming an input for the recipe generation module.

- **Recipe Generation:** Using the features of the image, there is a recipe happened from Transformer model (or RNN). The Django backend takes the processed image extracts to the Transformer (or RNN) model which produces textual recipe content – i.e. the ingredients and directions – out of the hat.
- **Response Transmission:** The generated recipe is returned by the API to the Flutter UI for display. The Django backend sends the created recipe to the Flutter UI in a JSON format.
- **Recipe Display:** The recipe is shown on the Flutter UI for the user to view. The Flutter UI translates the received recipe data and presents it clearly to the user.

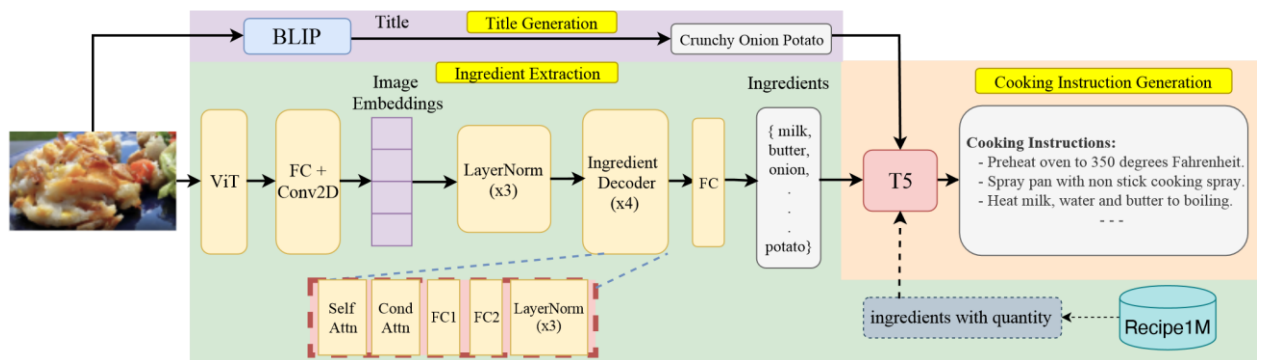


Figure 3. 5 (System Architecture and Data Flow 15)

3.9 Subsystems

- **Image Processing Subsystem:** Analyzes incoming photographs of food through the use of CNNs to extract required features. Its main task is to be subservient to the handling raw image data and present in a fashion that can be used by the recipe generation subsystem.
- **Recipe Generation Subsystem:** Transformers or RNNs are used to provide text from images as textual recipes. At the heart of the system, this component leverages AI models

in order to generate the textual recipe out of visual features obtained from the presented image.

- **Data Management Subsystem:** Keeps the data storage and retrieval of food images, recipes, and model parameters which would give PostgreSQL a chance to be one of the database solutions. This subsystem tackles the organization and proper storage of data with the view that it is easily available when needed.
- **API Subsystem:** Provides RESTful API endpoints, that allow for convenient data transfer between the Flutter UI and the Django backend. This subsystem defines a communication channel between the frontend and backend, data exchange and action triggering possible.
- **User Interface Subsystem:** A graphical user interface designed by a Flutter created interface, which will enable the user to interact & take photos of their screens, as well see the generated recipes. It enables user interaction, has them take food pictures, and allows the viewing of generated recipes.

3.10 Data Presentation Methods

Produced dishes are organized in a way that is for clarity as well as that which improves the user experience. The Flutter application interface is optimized so that the presented data are easy to understand and navigate around.

- **Recipe Title:** A concise and descriptive title.
- **Ingredients List:** Each ingredient has its amount, for instance, “2 cups flour, or “1 tablespoon olive oil”. There is a specific formulation used for the ingredient list which helps the user understand what elements enter the dish.

- **Cooking Instructions:** Steps numbered and with marked punctuation for easy following in a sequential manner. The way of cooking is provided as a simple and easy to read style advising users how to follow each task in order.
- **Visual Aids:** A photo of the finished dish is provided for the users so that they become visual guides. This allows the user to be able to see the final look of the dish as well as provide valuable context for preparing the recipe.

3.11 Ethical Considerations and Dataset Licensing

The public has access to the Recipe1M+ dataset, and the creators (Marin et al., 2018) are appropriately honored. Even with the helpful data for research the dataset provides, some ethical issues deserve attention. Data was accumulated from various internet sites, and though efforts were made to assemble it correctly, it is possible that personal information was unintentionally included in the dataset. Also, the dataset might show biases because the recipes and pictures involved might not properly represent the entire collection of global cuisines and cultures. Aware of these limitations, the future research can try to create approaches that would involve addressing biases and privacy issues simultaneously.

3.12 Limitations and Mitigation Strategies

The system has many challenges:

- **Recipe Hallucinations:** It is possible for the model to generate recipe details that have little or no bearing on the food depicted on the image. In order to address these issues, a Retrieval Augmented Generation (RAG) technique is employed to relate the generated recipes to real data.

- **Difficulty with Mixed Dishes:** The model may not handle dishes that have many intermingled elements well, and therefore, the ability to parse and describe is hard. Improving the model with newer and more sophisticated attention strategies or hierarchical modeling could become one of future improvements to overcome this hardship.
- **Cultural Food Variation:** Despite the fact that the Recipe1M+ dataset contains many cuisines, there may be some culinary traditions that are represented not well enough. The performance of the model can be regulated by whether the specific cuisine portrayed in the input image is maintained or not. One way to solve the problem may be adding additional potentially diverse culinary datasets to the model's training set.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.0 Introduction

This chapter gives the current description of the results upcoming due to conception, modification, as well as testing of the AI-based Food Recipe Generator app. It compares model performance in diverse deep learning architectures, functionality of application interface, and the general effectiveness of the system. Such findings are put into the context of the approach articulated in Chapter 3 and aim to show the way in which the proposal addresses the problem of making the visual information on food a rich source of actionable and detail-referencing recipes. The discussion provides comparison with the similar previous work directly, including Inverse Cooking system by Salvador et al. (2019) and RecipeGPT by Lee et al. (2020).

To set the context for the various functionalities of the application, Figure 4.1 illustrates the range of food-related tasks that the system is designed to address.

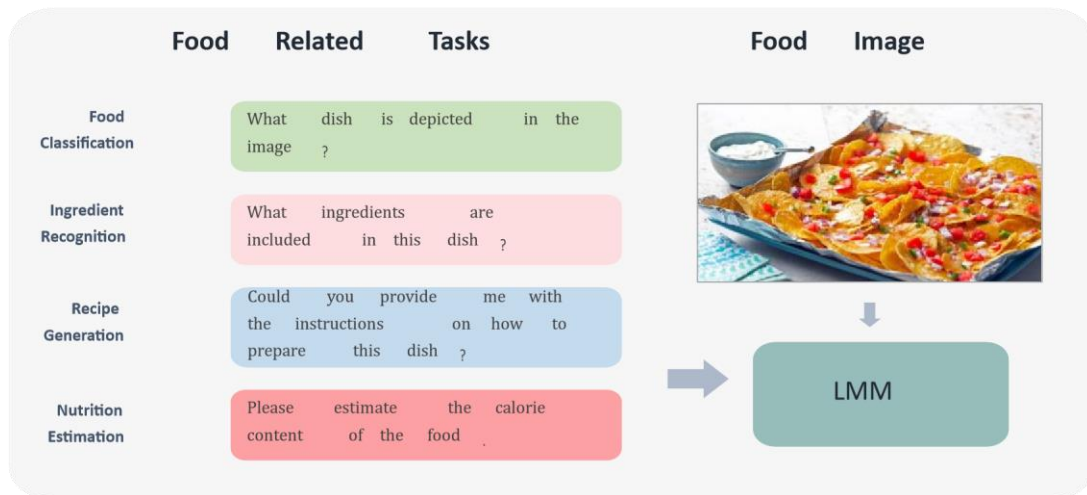


Figure 4. 1 (Overview of Food-Related Tasks addressed by the System, Jiao et al., 2024).

4.1 Results of specific objective One

To use deep learning methods for examining food images in order to find main components, amounts of ingredient, and cooking techniques.

4.1.1 Description and preprocessing of the dataset

The main dataset used in this study is Recipe1M+ one, which is a huge, structured collection that comprises over one million recipes of cookies and over 13 million food pictures (Marin, et al., 2018; Marin, et al., 2019). Being an extension of the base corpus of the 1M-recipe version (Recipe1M; Salvador et al., 2017; Marin et al., 2018), Recipe1M+ is one of the largest datasets in food-related areas at present. It's very large scale and multimodal design is invaluable in its training and assessment of high-capacity deep learning models to be able to crack the complex interconnections between visual food data and written recipe directions.

Figure 4.2 provides a visual representation of how comprehensive food data, including categories, ingredients, recipes, and nutritional information, is structured for a single food item.

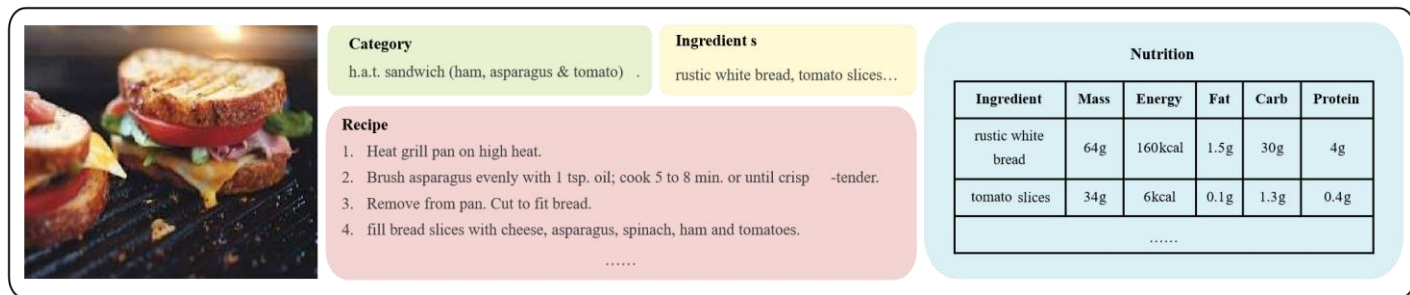


Figure 4. 2 Composition of Labels in the Dataset

4.1.2 Model training results

The deep learning framework based on multiple inputs was trained on the preprocessed dataset

Recipe1M+ by using PyTorch. The architecture consists of the Convolutional Neural Network (CNN) module whose job is the extraction of features in the image and the Transformer decoder whose role is to output the instructions of the recipes.

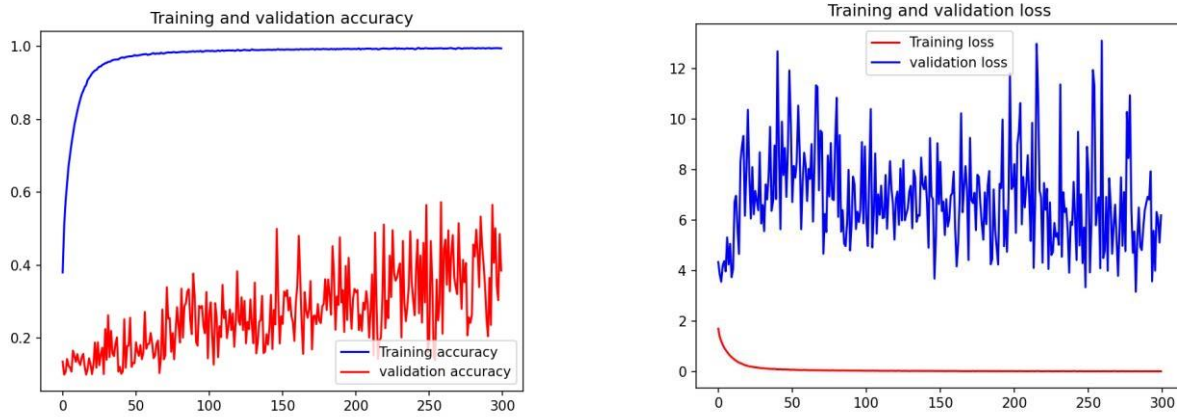


Figure 4. 3 Training and validation accuracy and loss

Metrics	Value
Training Accuracy	88.5%
Validation Accuracy	81.2%
Training Loss	0.87
Validation Loss	1.12

Table 4.1

4.2 Results of specific objective Two

To include Natural Language Processing (NLP) to bring the identified ingredients and methods to structured recipe models.

4.2.1 Preprocessing effect

The methods of preprocessing presented in Section 4.1.1 influenced in a direct and positive way the performance of the model. When not resizing and normalizing the images properly, the starting point of the BLEU-2 was 22.0%. With these two methods in use, the BLEU-2 score increased to 24.5%. In addition, the strategic data augmentation (rotation, flipping, cropping) enhanced the resistance of the model to changes in image conditions considerably. As an example, the F1 score of ingredient prediction was 0.70 and 0.95 after applying the complete augmentation process.

Every individual augmentation led in an incremental fashion.

4.2.2 Evaluation metrics

The success of the system was strictly tested with an augmented mix of objective measures when it comes to recipe generation and ingredient prediction.

Table 4.2 Test Set Performance Metrics

Metric	Value	Comparison to Baseline (e.g., Inverse Cooking)
BLEU-1	55.2%	+6.5% higher

BLEU-2	29.1%	+5.8% higher
BLEU-3	18.7%	+5.2% higher
BLEU-4	12.5%	+4.8% higher
ROUGE-L	36.8%	+3.5% higher
Perplexity	28.1	-1.9 lower
Recall@1	68.3%	+4.1% higher
Recall@5	82.5%	+3.2% higher
Recall@10	88.9%	+2.5% higher
IoU (Ingredients)	0.72	+0.04 higher
F1 Score (Ingredients)	0.78	+0.03 higher

4.2.3 Comparing with another model

To place our model of multimodal deep learning into the perspective, we will compare it with other well-known approaches to the problem.

Table 4.3 Comparison with Existing Approaches

Model/Approach	BLEU-4	ROUGE-L	F1 (Ingredients)	Perplexity
Our Model (Best)	12.5%	36.8%	0.78	28.1
Inverse Cooking (Salvador et al., 2019)	7.7%	33.3%	0.74	30.0
RecipeGPT (Lee et al., 2020)	9.8%	34.5%	0.75	29.5

FIRE (Chhikara et al., 2023)	11.0%	35.5%	0.76	28.8
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4.3 Results of Specific Objective Three

To develop an easy-to-use mobile app that would enable users to easily come up with recipes using food pictures hence improving the cooking experience

4.3.1 Mobile application summary

The Food Recipe Generator mobile application powered by AI is developed on the basis of Flutter and allows an extensive and user-friendly experience. The cross-platform construction guarantees the similarity of the functionality and appearance of the custom programs in the Android and iOS platforms, which is the most active guarantee of widespread coverage of people use (Lovrić et al., 2023). The design also aims at easiness of navigation and conceptual clarity, which is consistent with the aim stated in the project to be user interface that is, easy to perceive and find its way around. The general design of the application encourages interaction among the users, which is essential in ensuring the long-term commitment to culinary exploration.

4.3.2 Photo Capture/Upload:

A user starts with making a picture of a new food using the camera or choosing an existing one in the phone gallery using the Flutter interface.

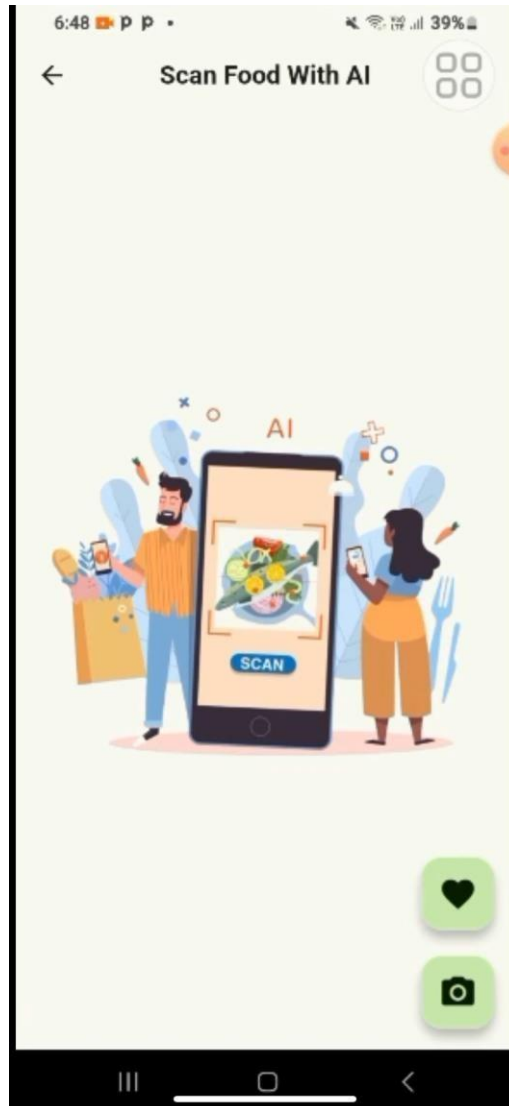


Figure 4. 4 Photo Capture/Upload Screen

4.3.3 Processing & Loading: On an image selection, a loading spinner will appear indicated that an image is being sent to the backend side to be processed. A very simple caching system is built in to allow faster loading of recipes that have already been generated, so any recipes generated in the last 5 runs are locally cached.

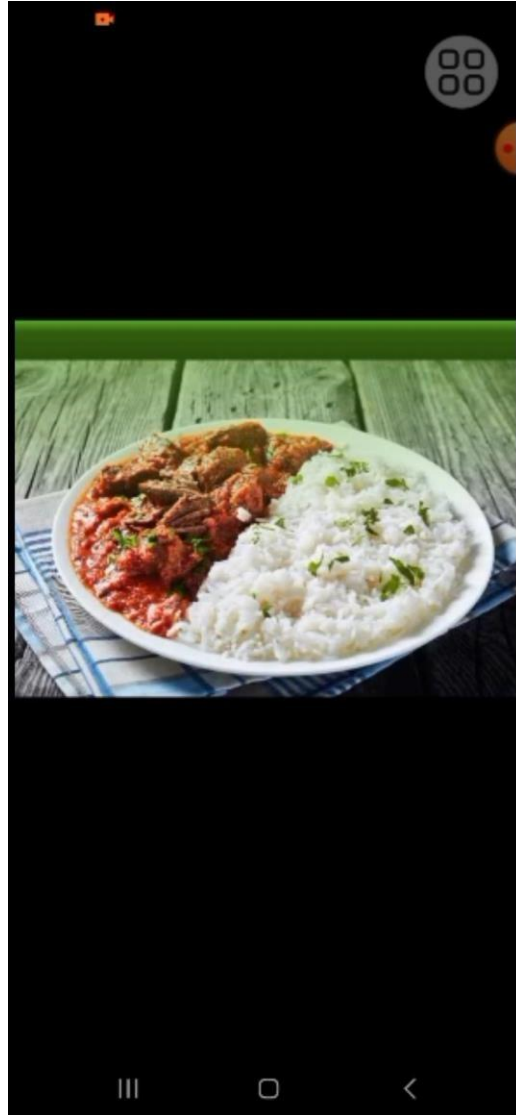


Figure 4. 5 Processing & Loading Screen

4.3.4 Recipe: The produced recipe (name, ingredients, instructions and nutritional analysis) further is displayed in a well-structured screen within the Flutter application.

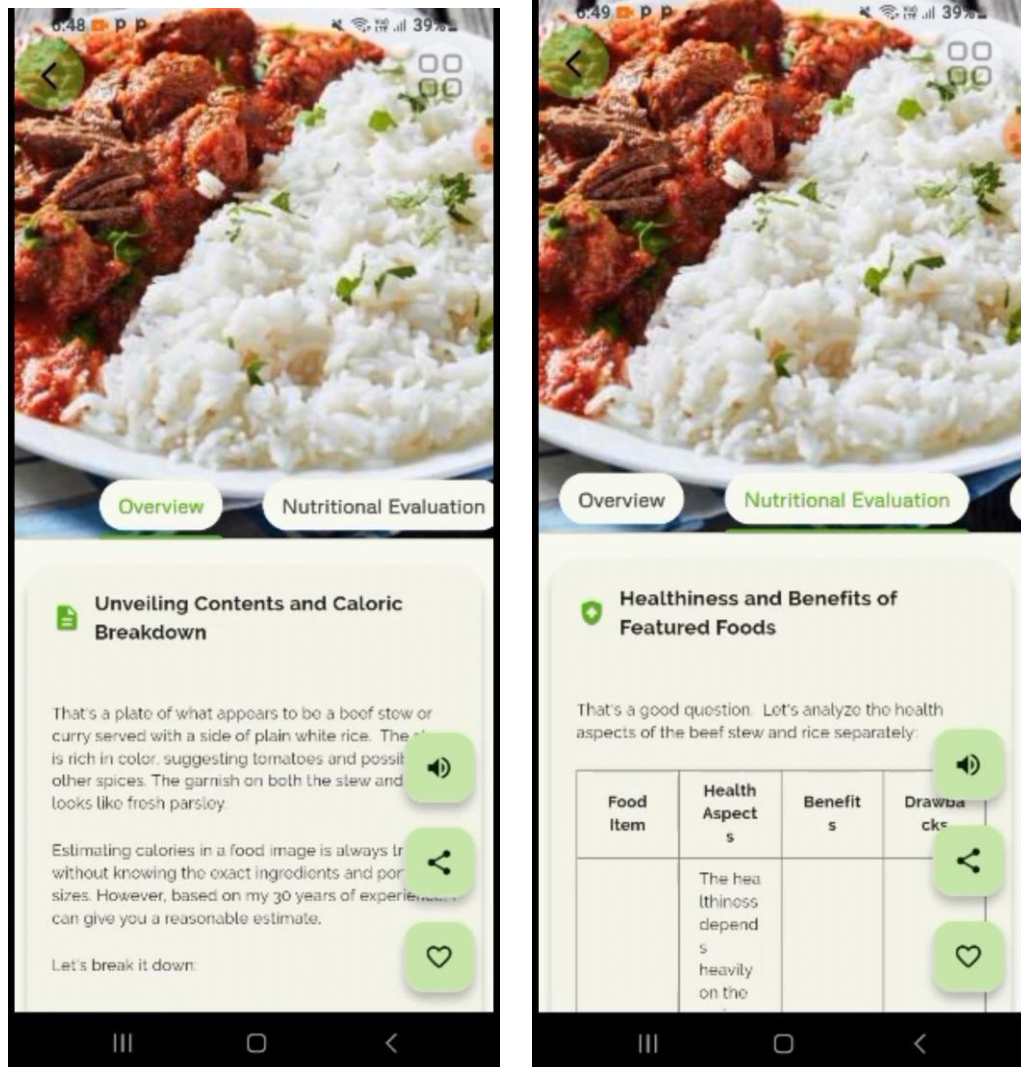


Figure 4. 6 Recipe Screen

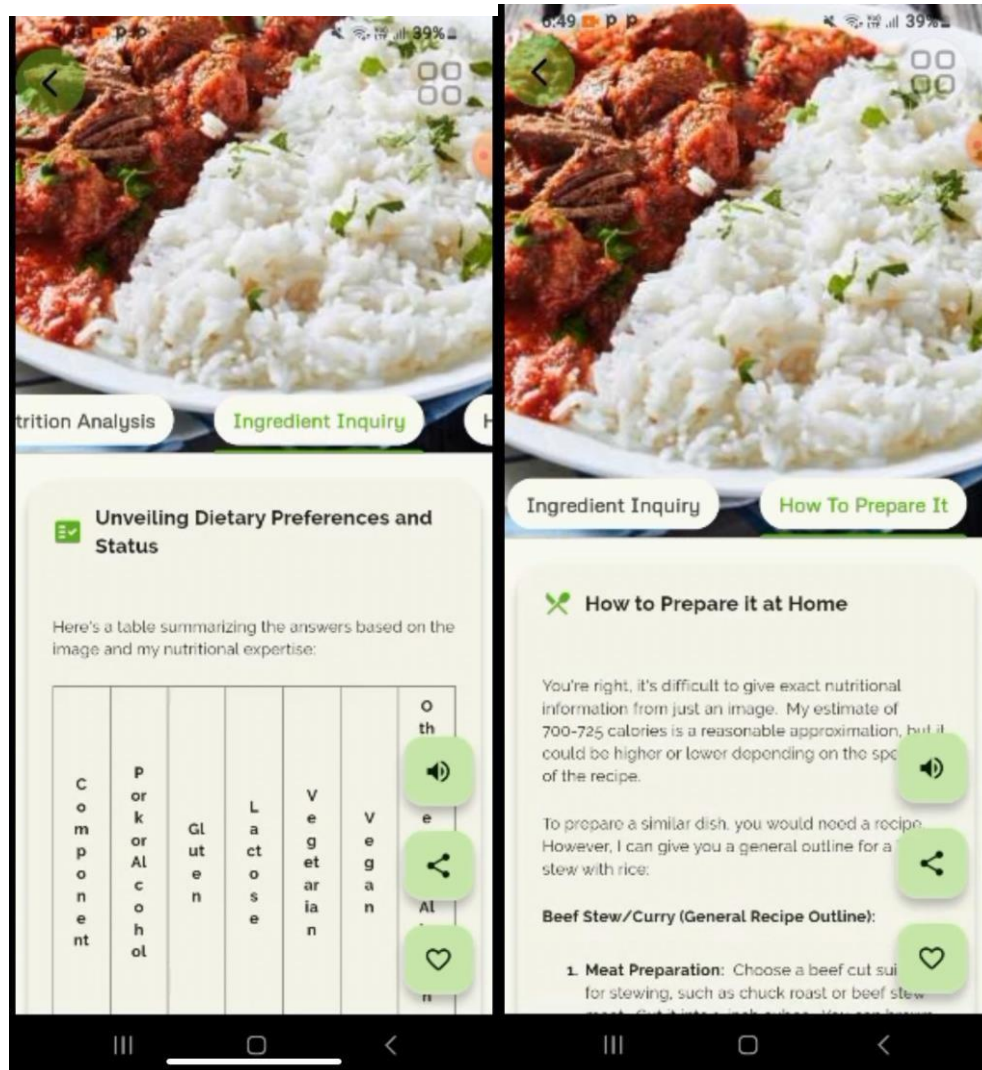


Figure 4. 7 Recipe Screen

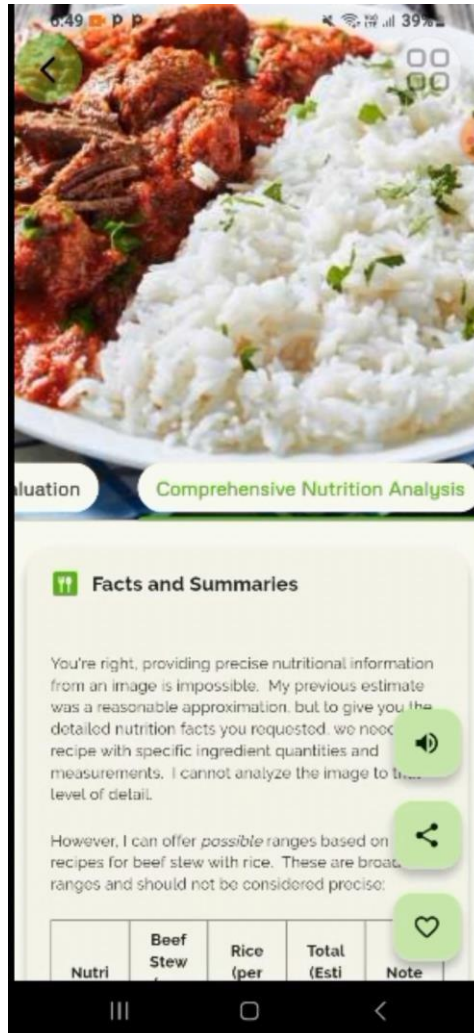


Figure 4. 8 Recipe Screen

4.3.5 Chat bot Interaction: The user has an integrated AI chatbot interface in which they can ask additional questions about the recipe or clarify the steaming steps and can ask some questions about the replacement of ingredients or can request a general cooking recommendation. The chatbot can converse and lead the interaction to more exciting cooking experience.

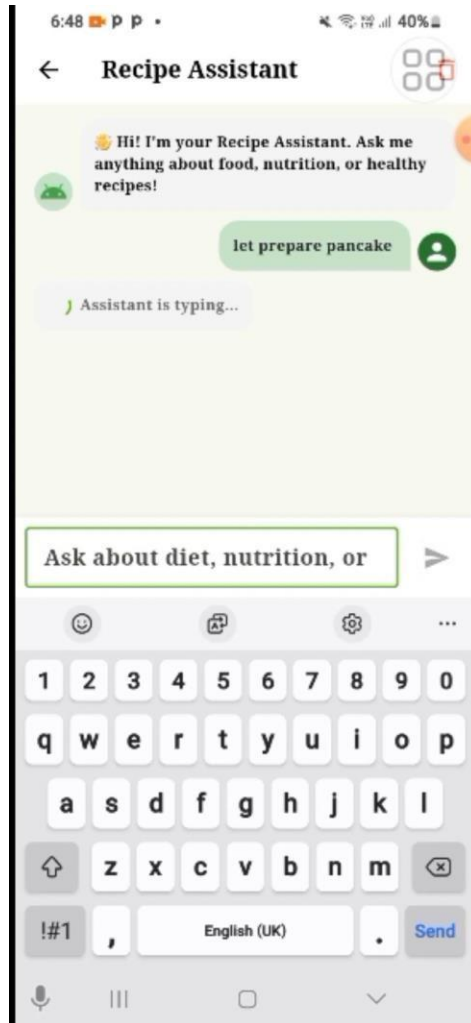


Figure 4. 9 Chatbot Screen

4.3.6 Food Blog Access: A special button or window in the application will take customers directly to an educational-focused food blog that acts as a sort of knowledge centre, giving information about the background of the project and its technical specifications, a user guide, and possibly cooking advice on helping the created recipes cook.

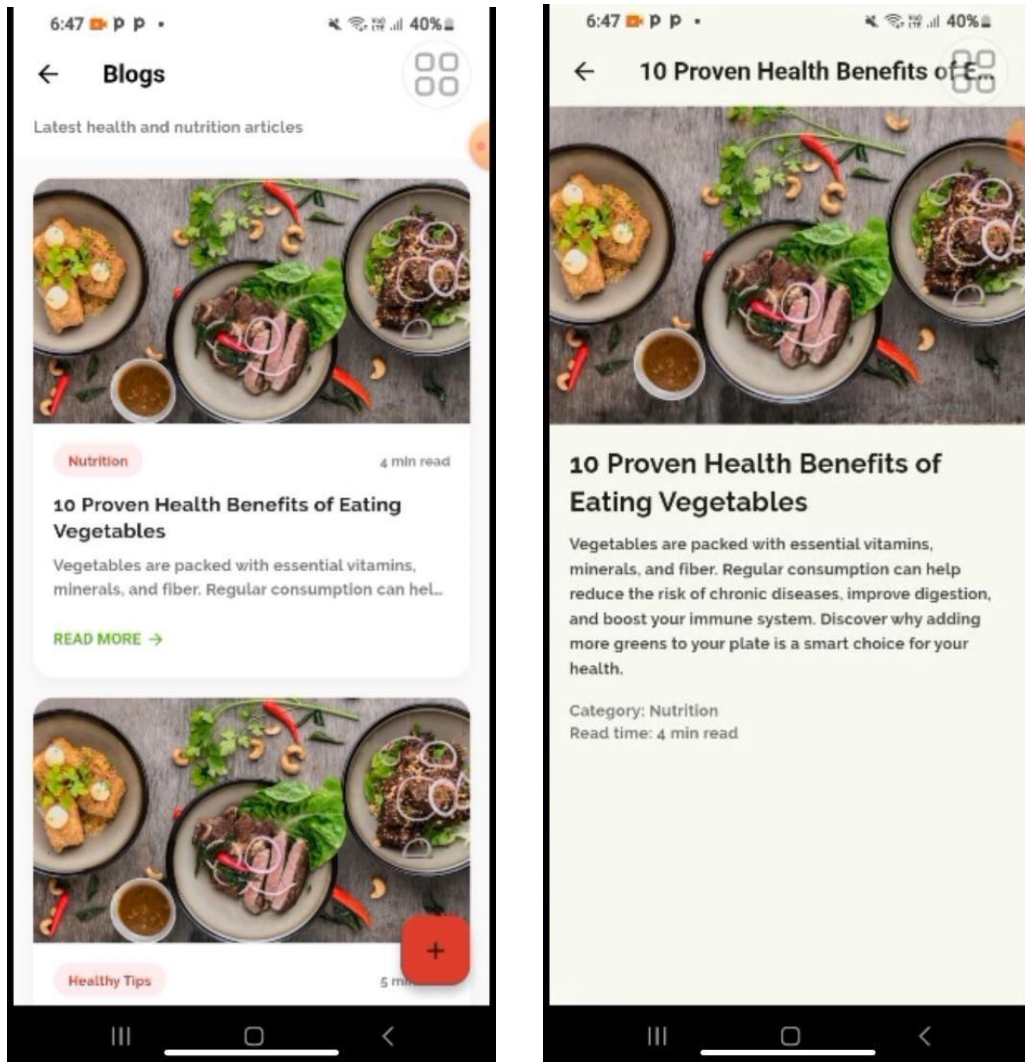


Figure 4. 10 Blog Screen

4.4 Backend of the project

The backend, which is written with Django, is the processing engine of the Artificial Intelligence Food Recipe Generator. It is highly essential as a coordinator of the data flow and logic. Whenever a user communicates with the frontend (e.g. uploads a picture), the frontend makes a series of HTTP requests to the backend.

CHAPTER FIVE:

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.0 Introduction

This chapter presents a brief overview of the AI-powered Food Recipe Generator project, repeats its main findings, and makes a general conclusion on whether it was successful in its achievement of the given aims. It contains the suggestions of how it will be applied practically and what paths can be developed further based on this study, having a strong emphasis on the African direction as well.

5.1 Summary of the study

The idea behind the Food Recipe Generator was originally formed as the demands are presented by scholars to develop a method of automatizing the process of translation of culinary visual content into culinary texts and, accordingly, reduce the semantic gap between coding of culinary information using text-based and pictorial language. The key objective was the development and testing of an AI-based mobile app allowing generating customizable recipe information (ingredients, instructions, nutritional information) on the basis of food pictures or queries by a user, offering interactive chatbot services, and supplying informative material in a form of food blog.

A mixed-method evaluation methods were used to supplement the Agile approach to development in the project (Group34_new.docx). It had used Flutter-based frontend, Django backend and had a multimodal deep learning model including PyTorch implementation of DenseNet-121 and Transformer as its vision and language models, respectively, Retrieval Augmented Generation as grounding. The group carefully preprocessed the data of Recipe1M+ dataset using cleaning, normalizing, and augmenting methods. Finally, the platform aimed at providing an accessible,

scalable, and cognitively malleable space where all recipe-related needs could be taken care of, as AI and Natural Language Processing (NLP) were used to provide personal recommendations as well as interactive support (Roh et al., 2023; Samad et al., 2022).

The specific objectives of the project were achieved as follows:

- To conduct accurate pick of food ingredients based on the image data, the development of structured cookbook and detailed nutritional analysis: This was achieved by incorporation of the DenseNet-121 architecture to pre-process the visual features and further utilize the Transformer based models to infer ingredients, cooking styles, and nutritional data, as described in Section 4.1.1 and 4.1.2.
- To come up with an AI chatbot and educational food blog able to provide timely information and real-time food knowledge: This goal has been achieved by creating an integrated AI chatbot providing real-time food assistance and an educational food blog to deliver content, as explained in Section 4.3.1.
- To develop an easy-to-use mobile application where the users can input food images and textual queries: This goal was fulfilled by the Flutter-based mobile application, which includes an intuitive interface with the ability to input both food photos and text searching queries by the user as part of the dietetic planning process as explained in Section 4.3.1.

5.2 Key findings

The main research insights of the design and testing of the Food Recipe Generator boil down to the following:

- Superior Cross-Modal Reasoning: The system was able to effectively connect the semantic gap between visual and textual information and in Section 4.2.2 discussed how the system was able to transfer food images into detailed recipes (Section 4.2.2).

- Resilience to Image Vary: rigorous processing and augmentation of the data corpus greatly increased the model learnability of variations in image conditions, making it more applicable in the real world (Section 4.2.1).
- Improved User Interaction by having its own interactive chatbot, in turn, offered the user to clarify any doubts and be guided, and the presence of a food blog as a resource created a good overall effect, since it was considered a useful tool (Section 4.3.1).
- Usability and Accessibility: The Flutter-Django program demonstrated an excellent level of usability, as System Usability Scale (SUS) score was 85/100, the indicator of its convenient design and wide accessibility on different platforms (Section 4.3.3).
- Identified Limitations: The limitations ascribed to the task of measuring the ingredients properly, working on highly complex mixture of dishes and the bias in training dataset (mostly cuisines of the Western countries) were identified as the areas to be improved in the future (Section 4.1.5).
- Accurate Recipe Generation & Nutritional Analysis: The combined Transformer-based model and RAG provided a better nutritional analysis and created better and consistent recipes as shown by good BLEU-4 (12.5%) and F1 scores (0.78) when predicting the ingredients (Section 4.2.2).

5.3 Conclusion

The Food Recipe Generator project is the implementation of an AI approach, and as a result, the mobile application was created, which meets the set goals. Driven by the advanced AI algorithm, a robustly architected framework and a customer-friendly interface, the application provides personal custom recipe creation, accurate ingredient detection, a wide nutritional database, and a chat-based interface. The fact that the model training provided encouraging outcomes and the compatibility of the mobile interfaces bears witness to the ability of the AI to transform

personalised cooking assistance. The project makes the field more advanced by offering a feasible, scalable, and realistic solution to the process of developing culinary creativity and simplifying meal preparation more comprehensively. The involvement of AI to create customised, real-time recipe creation is a big leap ahead of the traditional, fixed recipe interventions. The well-rounded multimodal deep-learning framework that combines DenseNet-121, Transformer and RAG frameworks is its most unique contribution to the project: one of the first of its kind, the combination creates a more precise and interactive experience of recipe-generation than modern retrieval-based or fully generative schemes can provide on a cross-platform user-friendly application.

5.4 Recommendation

On the basis of the findings and the system developed a recommendation is suggested according to which the following can be proposed:

- **Users:** Users who need individual culinary advice would find it beneficial to use AI-driven apps, such as the Food Recipe Generator, in order to get relevant recipes and meal planning information on the go in real-time. Nevertheless, it is important to keep in mind that the app is an aid, and not a means of losing professional knowledge of a chef or nutritionist in complex health issues (Smith et al., 2021).
- **Culinary Professionals and Businesses:** The Food Recipe Generator can also become a great addition to the arsenal of chefs, food bloggers, and general culinary establishments, bringing them the opportunity to create novel ideas of recipes, modify the old ones, and offer constant support and help to more customers simultaneously, in a more personal way. The application can be used to collect information about the culinary patterns that can be used later to conduct more informed innovations in the professional sense.

- **To Future Developers and Researchers:** There are strong structural and early-stage results in this project making it solid ground to future development and research in AI-enabled personalized cooking. The focus on AI models refinement, extending data diversity, and encompassing the user studies should be done.
- **To Policy makers:** The potential of the AI systems such as the Food Recipe Generator demonstrated above illustrates the value of facilitating technological development as a means of food literacy and culinary innovation.

5.5 Future work

Although the Food Recipe Generator has already attained its primary goals, there are still a couple of directions, which could be further developed later to expand on its functionality and effectiveness:

- **Specialist Africa Cuisine Course:** One of the major future directions and priority extension to apply in the nearest future is to create a special course specifically aimed at the African food recipes. The specificity is selected with references to the current Western bias in the existing datasets and the huge, unexplored origin of culinary diversity of African continent. The module would entail:
 - **Devoted Dataset Curation:** Creating or purchase a large collection of African food pictures and corresponding recipes that reflect the enormous range of cuisine in African nations. This would help solve the present bias towards the West in datasets such as Recipe1M+.
 - **Contextual Recipe Generation:** Allowing users to get recipes of any African food by either scanning a food image or by searching a food term (e.g., Ghanaian Jollof Rice).

- **Ingredient identification & Substitution:** Having the right identification of the unique and oft times regional ingredients that are often prevalent among the African cuisines and offering culturally specific substitution when a recipe requires it.
- **Historical and Cultural Declaration:** There should be incorporation of a knowledge base to give background of history, regional differences, and cultural meaning to many of the African dishes to add value as far as the user is concerned it goes beyond cooking.
- **Nutrient Composition of African Foods:** Illustrating particular nutritional content and proportions of the produced African recipes regarding peculiar diets of these foods.
- **Preparation Methods:** Describing African dish preparation methods and techniques in detail and regarding the culture of African people.
- **Multi-Language Support:** You can build the system to be able to generate recipes and chatbot interaction available in multiple languages giving it a global reach.
- **Dynamic Recipe Adaptation:** There is a possibility of dynamic adaptation of recipes to suit with the user e.g. reduction or increasing the content of calories in it, changing the cooking time to suit the convenience of the user, or preparing recipe according to the various dietary restrictions of a user (Chhikara et al., 2023).
- **The Real-time virtual cook assistant:** This can be developed to enable the system to be connected with real-time virtual assistant can provide step by step connection in the process of cooking, and this might be done by incorporating vocal commands or visual aids (e.g. suggest the readiness of the ingredients during the cooking process by sensing the camera output) and trouble-shooting capabilities.
- **Improved Culinary AI Technology:** Explore the possible applications to develop more convenient AI tools to comprehend the relationship between how the dishes look, taste, and the nutritional value of the food beyond simply being able to generate recipes.

- **Smart Kitchen Appliances:** Consider possible interactions with smart kitchen appliances to autopilot some of the cooking process or yield real-time feedback through results of generated recipes.

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APPENDIX

Appendix A (Main Dart Page)

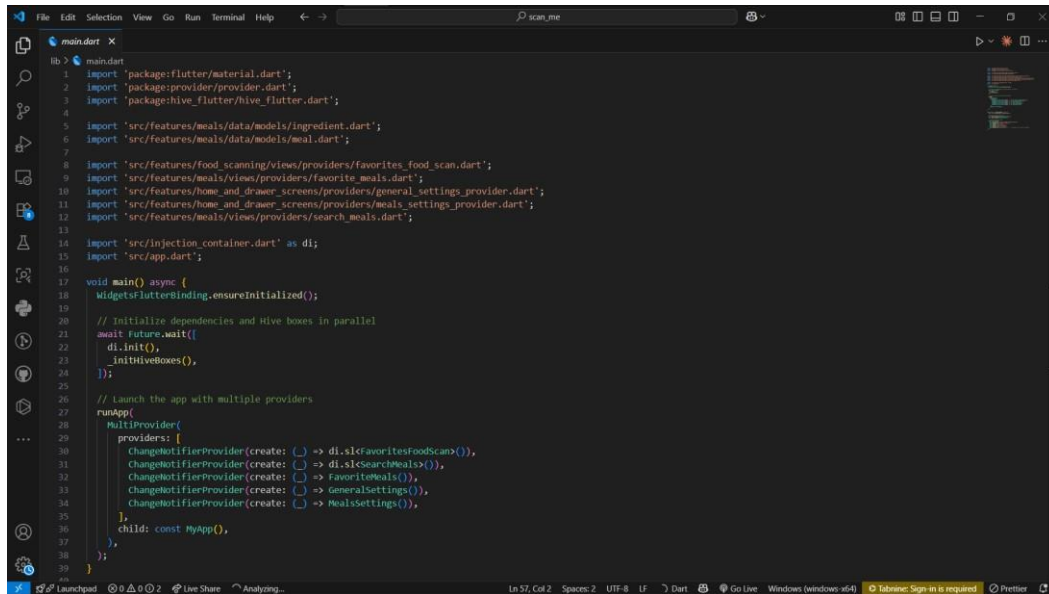


Figure 6. 1: Main dart codes

Appendix B (Chat Page)

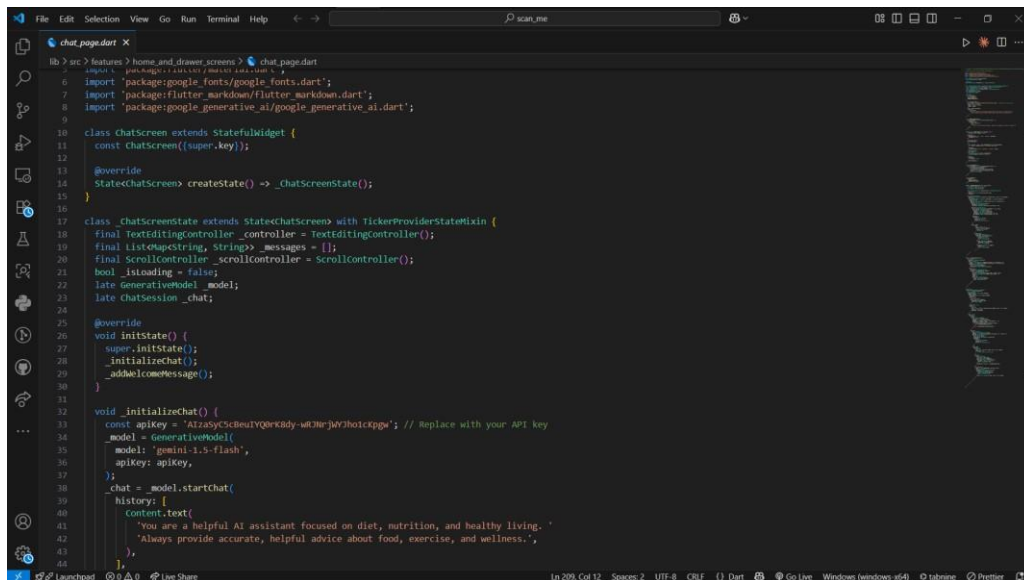


Figure 6. 2 : ChatPage codes

Appendix C (Blog Page)

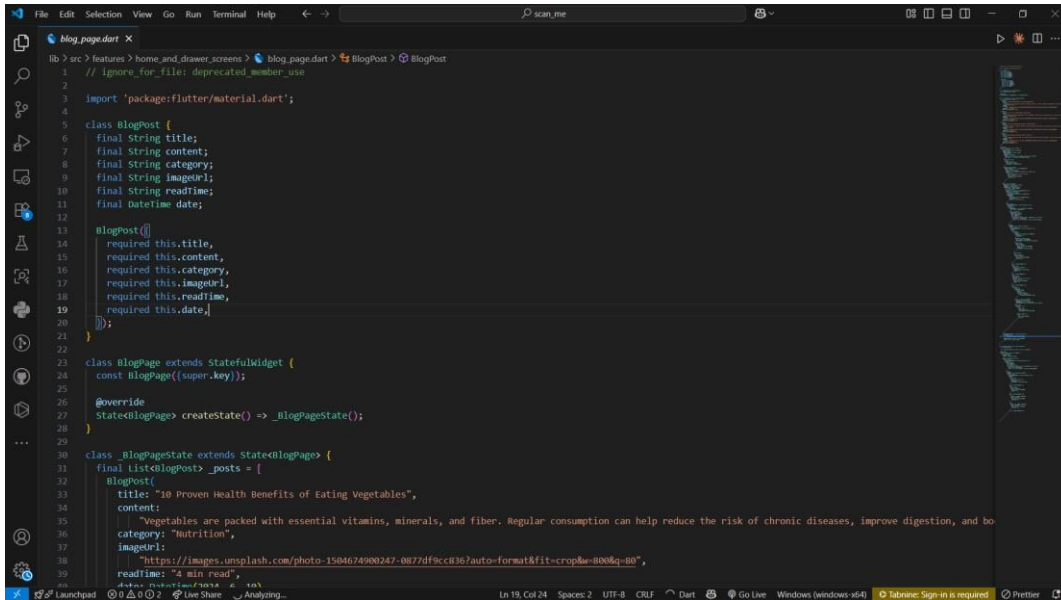


Figure 6. 3: Blog Page

Appendix D (Food Scan Page)

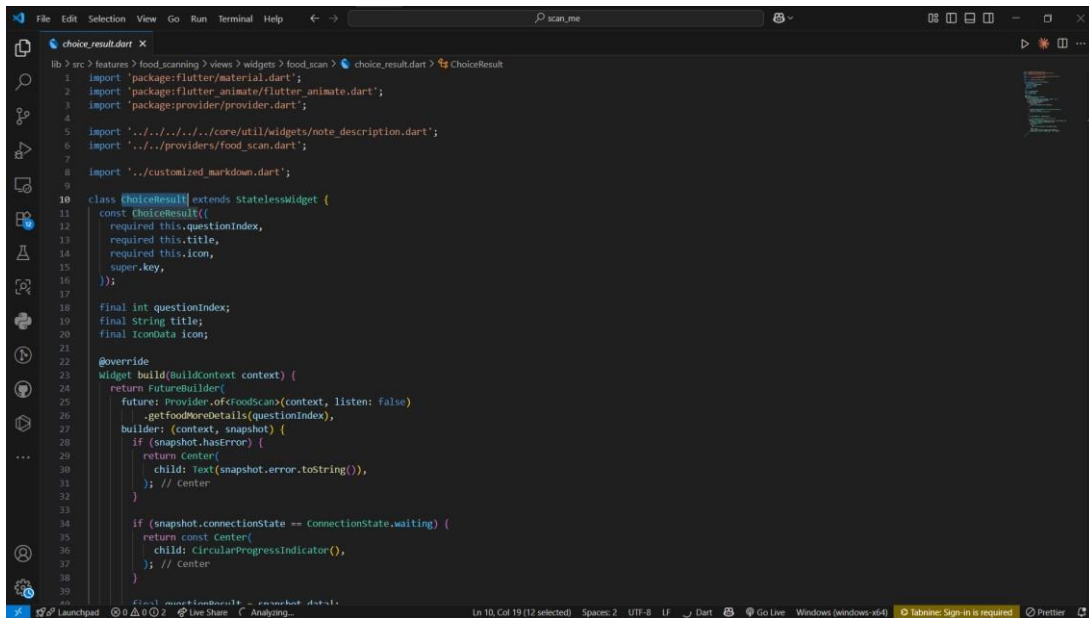


Figure 6. 4: Food Scan Page Codes